



Meeting of the Board
30 September – 2 October 2017
Cairo, Arab Republic of Egypt
Provisional agenda item 14(g)

GCF/B.18/04/Add.07

11 September 2017

Consideration of funding proposals – Addendum VII

Funding proposal package for FP052

Summary

This addendum contains the following three parts:

- a) A funding proposal summary titled “Sustainable and Climate Resilient Connectivity for Nauru”;
- b) No-objection letters issued by the national designated authority(ies) or focal point(s); and
- c) Environmental and social report(s) disclosure;



Funding Proposal

Version 1.1

The Green Climate Fund (GCF) is seeking high-quality funding proposals.

Accredited entities are expected to develop their funding proposals, in close consultation with the relevant national designated authority, with due consideration of the GCF's Investment Framework and Results Management Framework. The funding proposals should demonstrate how the proposed projects or programmes will perform against the investment criteria and achieve part or all of the strategic impact results.

Project/Programme Title: Sustainable and Climate Resilient Connectivity for Nauru

Country/Region: Nauru

Accredited Entity: Asian Development Bank

Date of Submission: 07 April 2017

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Section G	RISK ASSESSMENT AND MANAGEMENT
Section H	RESULTS MONITORING AND REPORTING
Section I	ANNEXES

A.1. Brief Project / Programme Information		
A.1.1. Project / programme title		Sustainable and Climate Resilient Connectivity for Nauru
A.1.2. Project or programme		Project
A.1.3. Country (ies) / region		Nauru
A.1.4. National designated authority (ies)		Department of Foreign Affairs and Trade
A.1.5. Accredited entity		Asian Development Bank
A.1.5.a. Access modality		☐ Direct ☒ International
A.1.6. Executing entity / beneficiary		Executing Entity: Ministry of Finance Beneficiary: Nauru Port Authority
A.1.7. Project size category (Total investment, million USD)		☐ Micro (≤ 10) ☐ Small ($10 < x \leq 50$) ☒ Medium ($50 < x \leq 250$) ☐ Large (> 250)
A.1.8. Mitigation / adaptation focus		☒ Mitigation ☒ Adaptation ☐ Cross-cutting
A.1.9. Date of submission		07/04/2017
A.1.10. Project contact details	Contact person, position	Pivithuru Indrawansa, Senior Project Officer (Infrastructure)
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A.1.11. Results areas <i>(mark all that apply)</i>	
Reduced emissions from:	
☐	Energy access and power generation (E.g. on-grid, micro-grid or off-grid solar, wind, geothermal, etc.)
☒	Low emission transport (E.g. high-speed rail, rapid bus system, etc.)
☐	Buildings, cities and industries and appliances (E.g. new and retrofitted energy-efficient buildings, energy-efficient equipment for companies and supply chain management, etc.)
☐	Forestry and land use (E.g. forest conservation and management, agroforestry, agricultural irrigation, water treatment and management, etc.)
Increased resilience of:	
☐	Most vulnerable people and communities (E.g. mitigation of operational risk associated with climate change – diversification of supply sources and supply chain management, relocation of manufacturing facilities and warehouses, etc.)
☐	Health and well-being, and food and water security (E.g. climate-resilient crops, efficient irrigation systems, etc.)
☒	Infrastructure and built environment (E.g. sea walls, resilient road networks, etc.)
☐	Ecosystem and ecosystem services

(E.g. ecosystem conservation and management, ecotourism, etc.)

A.2. Project / Programme Executive Summary (max 300 words)

Nauru is the world's smallest island nation. Isolated deep in the Pacific, it depends almost entirely on its port for supplies of food, energy, and most other essentials its people and economy need to survive. Yet its aging dysfunctional port facilities are completely exposed to the effects of climate change.

Aiwo Port's century-old design leaves visiting ships totally unprotected from sea and climatic conditions. The offshore loading and unloading necessary has become steadily more hazardous and difficult. Delays and complete port shutdowns are chronic for 3 months a year, and shipping costs are climbing. Some ships already refuse to serve the port.

Westerly waves and swells that pose the main challenges at Aiwo are driven by the El Nino Southern Oscillation, the South Pacific Convergence Zone, the Western Pacific Monsoon, and distant tropical cyclonic activity. The great uncertainty about how climate change will affect these drivers raises the significant risk that the wave climate at the port will worsen and combine with sea level rise to make Nauru's vital sea link inoperable for longer periods every year.

The alternative proposed for GCF's support is construction of a climate-resilient port that can operate year-round. Of five designs appraised, the one selected comprises (i) a channel through which oceangoing ships can pass between the sea and the shore for the first time in Nauru's history, (ii) a stable wharf with a turning berth, (iii) a breakwater to shelter the wharf and the berth from waves, and (iv) port buildings, container terminal and port security provisions complying United Nation conventions such as International Ship and Port Facility Security (ISPS) Code. This project will also help reform port governance and management and capacity building of Nauru Port Authority (NPA) to ensure financial, economic, and institutional sustainability.

The new port facilities will mean that ships spend considerably less time at sea, leading to significant reduction in CO₂ emissions from the shipping sector.

Impact potential: All Nauru's 11,300 people depend in multiple ways on the country's sea link. Making the port fully functional despite climate change will benefit them all. Over the 50 years of the port's lifetime, estimated reductions in CO₂ emissions are 535,400 tons.

Paradigm shift potential: Ports are at the forefront of infrastructure at risk of climate change and the project can provide replicable solutions for the many Pacific islands that face similar challenges. Reliable maritime connectivity can transform Nauru's society, economy, and future.

Sustainable development potential, and needs of the recipient: Making the port a lifeline, rather than a choke point, will create new opportunities for development, employment, and poverty reduction; bring down high costs of such basics as food and water; and improve public health through dependable flows of nutritious food and medical supplies. Education will benefit. The gender action plan (GAP) promotes benefits for women.

Country ownership. Government leaders, including the National Designated Authority (NDA), consider the project a priority. It aligns with several national policies and plans, and the design is based on thorough consultation. Refer Annex 1A for a 'no objection' letter from the NDA and Annex 4 for commitment letters from the Government of Nauru and Government of Australia confirming their financial contributions to the project.

Efficiency and effectiveness. The ratio of benefits returned for GCF's support will be high for GCF investments in the Pacific region. Several development partners are contributing. GCF's contribution covers 41% of total project costs, or an estimated 73% of costs eligible for climate change adaptation.

A.3. Project/Programme Milestone

Expected approval from accredited entity's Board (if applicable)

September 2017

Expected financial close (if applicable)	30 June 2023
Estimated implementation start and end date	Start: 27/11/2017 End: 31/12/2022
Project/programme lifespan	Implementation period: 5 years (construction completion by 2019; port reform implementation until 2022) Benefit stream: 50 years

B.1. Description of Financial Elements of the Project / Programme

Grant financing is requested from the GCF to ensure that Nauru continues to have maritime transport connections to the region and the world. The project's financing plan is provided in Table 1. It shows that ADB, the Accredited Entity, is partnering with the Government of Nauru and the Government of Australia on this project. The Government of Nauru is contributing counterpart funds of US\$4.09m for the project, plus an in-kind contribution in the form of counterpart staff, office accommodation and audit costs valued at approximately US\$6.27m, and exempting duties and import taxes. The Government of Australia is providing grant financing of US\$13.5 between FY2018–FY2020¹. Refer Annex 4 for confirmation/commitment letters from the governments of Nauru and Australia.

Table 1: Project financing plan

Source	Amount (US\$ million)	Percentage
Asian Development Bank	20.70	32%
Government of Australia	13.50	21%
Government of Nauru ²	4.09	6%
Green Climate Fund	26.91	41%
Total Project Cost	65.20³	100%

Nauru is a small island developing state (SIDS) and highly vulnerable to external economic shock and climate change. Its small size, remoteness from partners and markets, and limited economic diversification opportunities mean it cannot access commercial credit on favourable terms. As a SIDS with high debt levels and high vulnerability to climate change, Nauru requests grant financing.

GCF funds are to contribute to the investment in the construction of a port in Nauru that is resilient to climate change. As part of this project but not supported by GCF, the Government of Nauru will introduce port reforms and capacity development of staff to enable the NPA to generate revenue and thereby make the port operations financially sustainable over the long term.

Full projections of expenditures, cash flow, and revenue for port operations until 2027 are provided in the financial analysis (Annex 2F). Overall project costs and GCF support for each component are provided in Table 2.

Table 2: Total project costs and GCF amount sought by component

Component	Total Amount (US\$ million)		Total Amount Local Currency (AU\$ million)		GCF Amount (US\$ million)	GCF Amount Local Currency (AU\$ milli on)	Currency of disbursement to recipient
Component 1: Priority port infrastructure	62.70		82.5		26.91	35.41	USD
Offshore works							
Berthing channel	7.84	10.32	4.60	6.05	USD		
Breakwater	6.47	8.51	3.80	4.99	USD		
Wharf	31.56	41.52	18.51	24.36	USD		
Onshore works							
Port buildings	5.83	7.67	0	0	USD		
Building Services	1.46	1.92	0	0	USD		
Port Equipment	6.76	8.90	0	0			

¹ FY before a calendar year denotes the year in which the fiscal year ends.

² Plus in-kind contribution by Government of Nauru of US\$6.27m.

³ This total project cost is exclusive of taxes.

Supervision Consultant Services	2.78	3.66	0	0	USD
Component 2: Reforms for the NPA	2.50	3.29	0	0	USD

Table 3: Project budget by expenditure item

Description	Total Amount (US\$ million)	GCF (US\$ million)	Others (US\$ million)
Civil works—offshore (wharf, berth pocket, breakwater)	45.87	26.91	18.96
Civil works—onshore (port buildings and services)	7.29	0	7.29
Port equipment	6.76	0	6.76
Consultant services for construction and civil works	2.78	0	2.78
Port reforms, governance, and capacity development	2.50	0	2.50
Total	65.20	26.91	38.29

B.2. Project Financing Information

	Financial Instrument	Amount	Currency	Tenor	Pricing		
(a) Total project financing	(a) = (b) + (c)	65.2	Options				
(b) GCF financing to recipient	(i) Senior loans		Options	() years	() %		
	(ii) Subordinated loans		Options	() years	() %		
	(iii) Equity		Options		() % IRR		
	(iv) Guarantees		Options				
	(v) Reimbursable grants *		Options				
	(vi) Grants *	26.91	million USD (\$)				
	Justification for grant is provided in Section F.1.						
Total requested (i+ii+iii+iv+v+vi)		26.91	million USD (\$)				
(c) Co-financing to recipient	Financial Instrument	Amount	Currency	Name of Institution	Tenor	Pricing	Seniority
	Grant	20.7	million USD (\$)	Asian Deve. Bank	() years	() %	Options
	Grant	13.5	million USD (\$)	Govt. of Australia	() years	() %	Options
	Grant	4.09	million USD (\$)	Govt. of Nauru	() years	() % IRR	Options

	Lead financing institution: Asian Development Bank
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B.3. Financial Markets Overview (if applicable)

Not Applicable

C.1. Strategic Context

Nauru is a remote, small island developing nation with a land area of approximately 21 square kilometres and a population of about 11,300⁴. Nauru is the world's smallest island nation and is located far from all international partners and markets. It lies 40 kilometres (km) south of the equator and about 4,000 km from Sydney, Australia. See Figure 1.

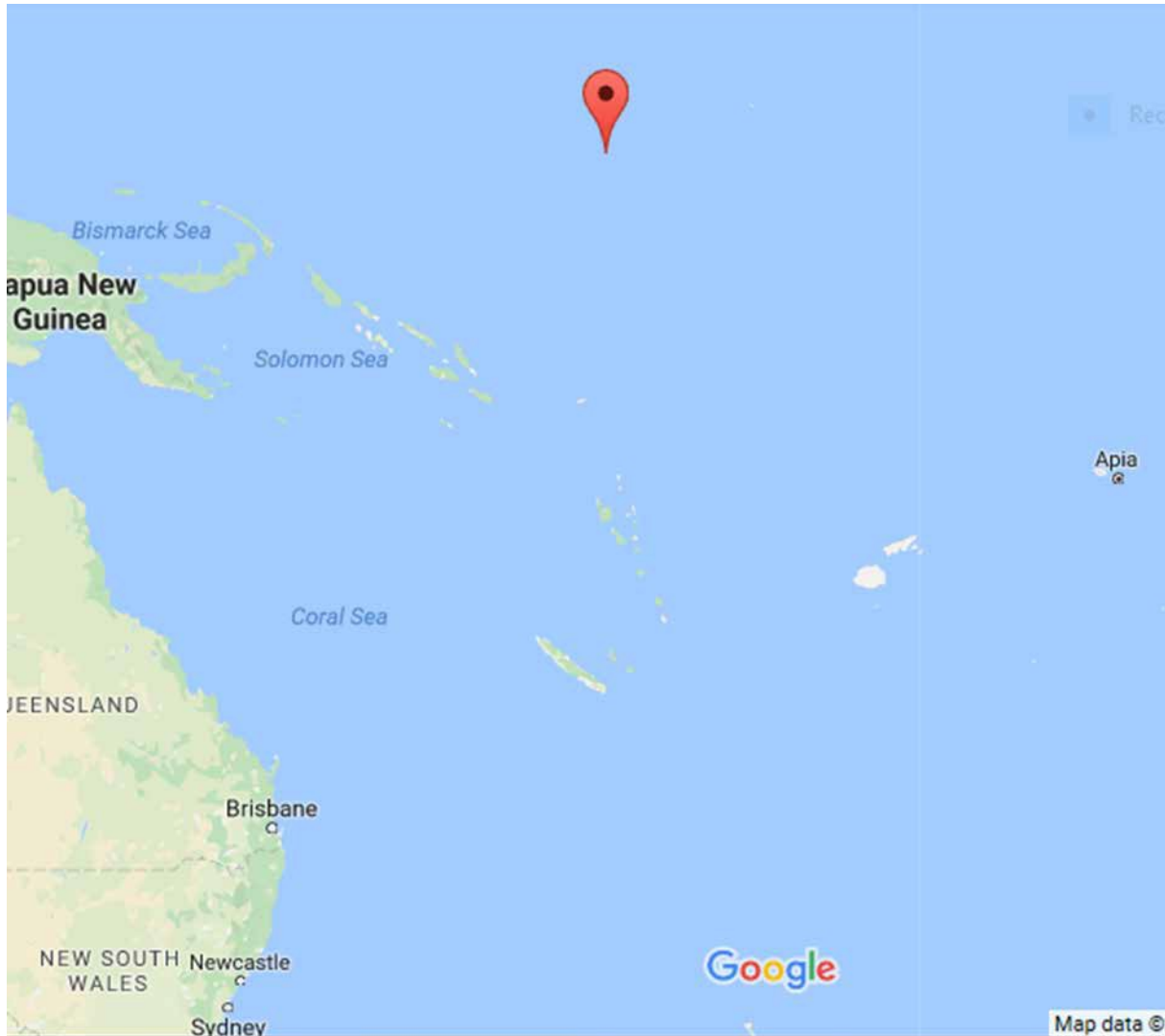


Figure 1: Map showing location of Nauru

The main economic activities in Nauru since the country's independence in 1968 have included the sales of fishing rights and some offshore banking and coconut processing. But mining and export of the island's once abundant phosphate deposits has long dominated the economy. Fluctuations in global commodity prices and the growing technical difficulty of extracting the remaining deposits have undermined the sector's productivity. The government has come to recognise that the nation's great dependence on this single industry is not a viable development strategy. Some of the latest available statistics provide an outline of the socioeconomic context in which the project will be undertaken:

- Gross domestic product per capita in 2014—US\$1,000
- Portion of the population living below the country's basic needs poverty line in 2013—24.0%

⁴ Source: MOF / National Bureau of Statistics website (2015 projection, based on 2011 census). This does not include approximately 1,200 residents of the Government's regional processing centre.

- Overall unemployment rate in 2011—22.9%
- Youth unemployment rate in 2011—more than 45.0%
- Life expectancy at birth in 2011—60.3 years
- Infant mortality as of 2011—33 per 1000 births.

Nauru is one of the smallest and most remote countries in the world. It has only a limited human resource pool and little economic diversity. Like all the Pacific island states, it is not only extremely exposed to the effects of climate and other natural hazards, but also lacks the capacity on its own to reduce its vulnerability to coming effects of climate change through adaptation. The Fifth IPCC Technical Assessment Report (IPCC AR5) reconfirmed the high vulnerability of small island states to climate variability and climate change.⁵

Nauru's isolated, distant location and almost total reliance on maritime transport makes its port a focal point of this vulnerability. Barring some passenger air traffic, all its economic, social, and cultural exchanges with the rest of the Pacific region and beyond are done by sea. Its current economy, any future economic development, food security, energy security, water security, basic health care, and education all rely almost exclusively on maritime transport. Of all the food its people consume, 90% must be brought in by ship and transit through the increasingly hazardous offshore operations at its delay-and shutdown-prone Aiwa port.⁶ So must 95% of the country's imports.

The lack of subsistence economy from their non-farming/gardening pushes the growing population in total dependency on imported goods and to increasing poverty. Consumer expenditure for food and non-alcoholic beverages consumed the biggest percentage at 43%. While the incidence of food poverty was assessed to be zero still 24% of the population are below the basic needs poverty line and 7.9% are extremely vulnerable. A reliable and climate proofed maritime transport will bring essential goods to the country more cost effectively and increase affordability by the population. Diet related non-communicable diseases is the primary cause of premature mortality without any improvement in the life expectancy levels in the country for a long time. This can also be potentially addressed through food imports of fresh and nutritious foods which can be better handled and transported given more reliable port operations. Further, expected increased in economic activities through improved maritime transport has the potential to bring a more diverse income opportunities. The government and state-owned enterprises are currently the major employer in the country and there should be enough room for private sector involvement including small enterprises.

C.2. Project / Programme Objective against Baseline

This section describes the current port facilities and operations and their growing vulnerability to climatic factors. It details the infrastructure improvements envisaged in the baseline scenario before explaining how projected climate change will present high risks to the baseline port facilities and operations even with the baseline infrastructure improvements. These risks may soon bring climatic conditions to a tipping point, at which the port will no longer be able to provide the efficient and reliable maritime transport Nauru is so dependent on. The section ends by introducing an alternative scenario, whereby climate resilient port facilities are constructed and they provide a sustainable, reliable basis for efficient maritime transport.

The Existing Port Facilities and Operations

Facilities

The current port facilities in Nauru have four main components.

1: A small, protected harbour on the Aiwo shore. Nauru's main port is located at Aiwo on the west coast, where a small, protected harbour was first built more than 115 years ago (Figure 2). Only small boats and flat barges can cross over the reef that surrounds the island and largely protects it from offshore waves. This has meant that despite several upgrades since it was built, the harbour has been used only by small boats, such as the barges and lighters that must currently ferry cargo back and forth between the harbour dock and oceangoing ships offshore. The onshore facilities at this harbour include basic access, storage and processing facilities, and the equipment used to load and unload cargo between the shore and the barges and lighters.

⁵ For example, see the report of AR5 Working Group II, Chapter 29 (Small Islands) (IPCC, 2013)

⁶ Source: FAO, 2016. See <http://www.fao.org/sids/resources/projects/detail/en/c/281992/>



Figure 2: Nauru and Aiwo harbour and project site

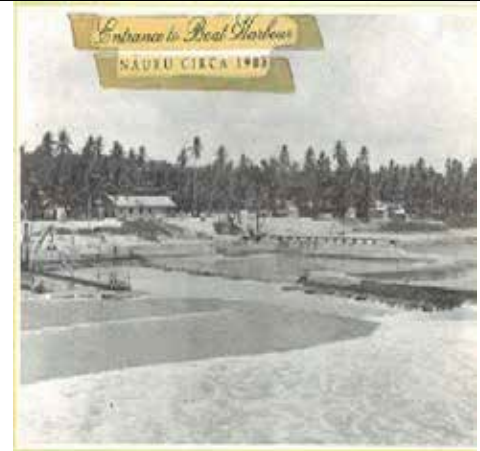


Figure 3: Aiwo harbour, 1908



Figure 4: Aiwo harbour, 2016

2: An offshore mooring system at Aiwo. Oceangoing cargo and passenger ships are unable to access the coast at Nauru due to shallow coastal waters and the high coral reefs. They are instead moored offshore for unloading and loading, using a mooring system first built to secure such vessels about 60 years ago. The system employs 540-meter-deep sea anchors, chains, connecting plates, shackles, and mooring buoys to hold and help manoeuvre vessels during these transfers.

3: Facilities for offshore loading and unloading at Aiwo. A system of cantilever arms is used to load phosphate from shore to oceangoing vessels offshore. The facilities include a hose system to discharge fuel from oceangoing ships positioned offshore.

4: A small fishing harbour at Anibare on the east coast. The harbour at Anibare was built for small fishing boats, not for transporting cargo. It has only a small wharf, very few facilities and little space for handling or storing cargo, and no mooring facilities offshore for oceangoing ships.

Operations

Maritime transport operations at Aiwo port are undertaken in three main ways. All are exposed to open ocean conditions, westerly winds and waves, and risks of interruption or shutdowns.

All cargo except phosphate and fuel. Apart from phosphate and fuel, all incoming and outgoing cargo must be transferred over the reef between the small shoreline harbour and oceangoing ships offshore by small, self-propelled,

flat-bottom barges. The ship–barge cargo and passenger transfers at sea must be executed using the ships' own cranes and gear (Figure 5). A mobile land crane handles the barge–shore transfers in the protected harbour.

Climatic challenges to offshore cargo handling. Sea conditions dictate the pace at which the ship–barge interchanges can be undertaken, as well as when they can be risked at all. In rough weather, the process is slow, complex, dangerous to workers and passengers, subject to delays, and often damaging to the cargo. When sea conditions become particularly difficult, cargo and container vessels cannot use the mooring system and must be left to drift. This makes any transfers more difficult and hazardous. (The mooring system is prone to damage and is currently inoperable.) The transits to and from the ships and the small harbour are also difficult and dangerous in heavy seas, and when conditions become too harsh they cannot be attempted.

Phosphate exports using mooring system. Oceangoing bulk phosphate carriers are moored at sea to the offshore mooring system prior to loading. The phosphate is transported by belt along the cantilever arms and deposited into a vessel's holds. The process requires that the ships be repositioned and pulled away from shore whenever the cantilever loaders must be moved for proper loading.

Fuel imports using mooring system. After a tanker is secured at sea using the port's mooring system, fuel is pumped through the port's flexible hose equipment to facilities on land.

Climatic challenges to use of mooring system for phosphate and fuel shipments. Phosphate bulk carrier and fuel tankers are left to drift when heavy weather makes the mooring system unusable. Loading and offloading in these conditions often requires the rental of specialized and expensive tugs from Solomon Islands to control this drift. With the mooring system currently damaged, these tugs are now needed to hold all such vessels in place while loading and discharging their cargoes (Figure 6).

Shifts to fishing harbour during Aiwo's shutdowns. When ocean conditions are extreme at Aiwo but calm at the small Anibare fishing harbour on the island's opposite coast, some limited loading and offloading operations are undertaken there.



Figure 5: A ship offloads a container on to a flat barge during calm conditions



Figure 6: Port facilities at Aiwo showing how the oceangoing tanker remains off-shore during unloading. The mooring buoys (near the tanker) are too small to be visible. However, as the buoys are currently damaged, the oceangoing tanker is held at sea by a tug - an operation that is only possible during calm conditions.

Climate Hazards Facing Aiwo Port Facilities and Operations

Because Nauru's main port of Aiwo is on the island's western coast, it is largely protected from easterly winds and waves but very exposed to westerly winds and waves. This section examines the climatic factors that either alone or in combination affect the facilities and operations at Aiwo port: westerly waves; westerly winds; sea level; and, potentially, rainfall and temperature.

Westerly waves. There are two sources of westerly waves: the westerly winds driven by the Western Pacific Monsoon (WPM), and tropical cyclones. Nauru does not experience tropical cyclones directly, since it lies close to the equator, but they regularly pass to the southwest. Even a distant cyclone can create ocean swells that cause strong, westerly swell waves to reach Nauru. IPCC AR5 provides several examples of storm swells causing extreme waves and flooding in the Pacific several thousand kilometres distant from the actual storm.⁷

The rose diagrams in Figure 7 illustrate the major role played by westerly winds and tropical cyclones in the wave climate offshore of Aiwo port. The diagram at the left shows that the clear majority of waves at Aiwo are easterly and small, below 1.75 m in height. The diagram at the right, which deals only with waves of more than 2 m, shows clearly that the clear majority of high waves are westerly. This is due to the WPM and tropical cyclones. A full analysis of the wind and wave speed and direction in recent years off the coast at Aiwo is provided in Annex 2N.

⁷ See Chapter 29 in IPCC (2013).

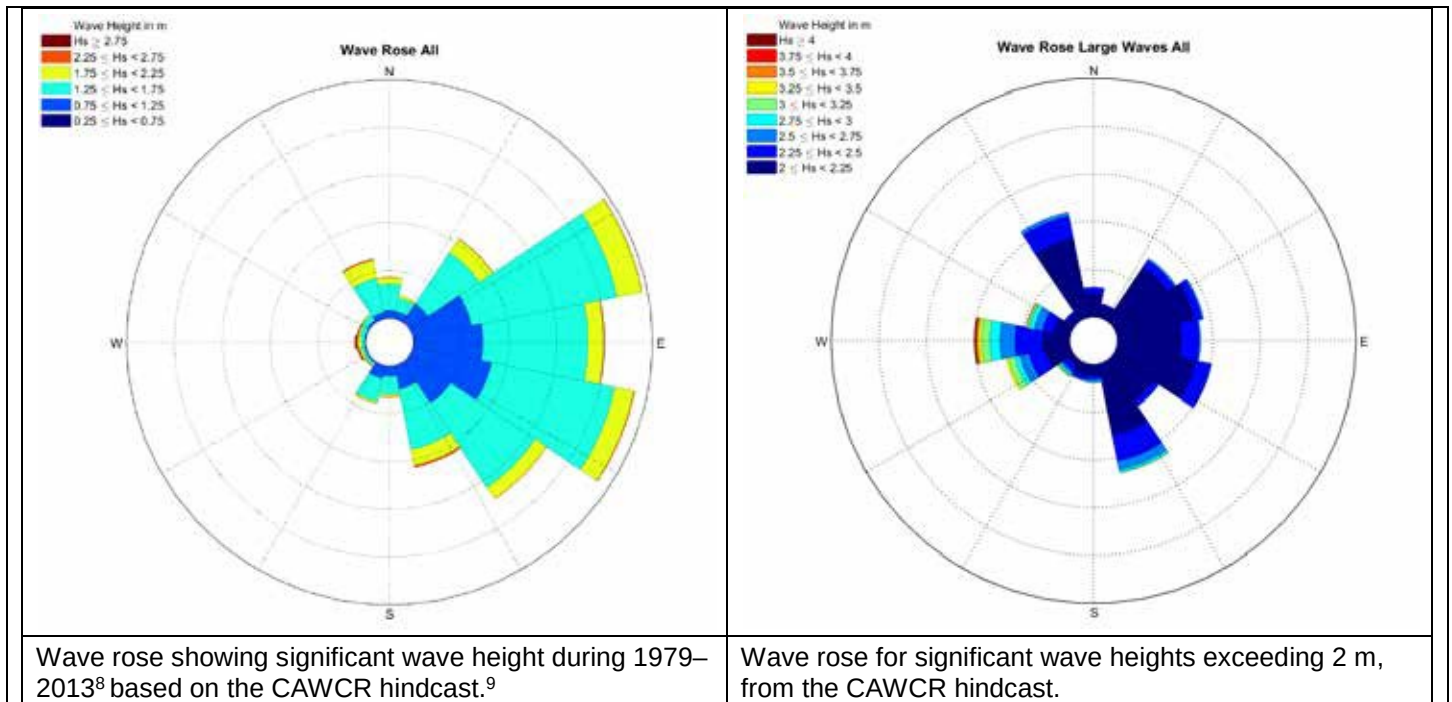


Figure 7: Wave roses for offshore at Aiwo

Westerly winds. Westerly winds are absent through most of the year and throughout some years. But westerly winds dominate when the WPM reaches Nauru. The forces driving the WPM are not fully known. However, it is believed that there is a strong relationship with the El Nino Southern Oscillation (ENSO) and the location of South Pacific Convergence Zone (SPZC).

Sea level. Seasonal and daily fluctuations in the sea level may affect the port's onshore and related facilities in two ways. High sea levels may lead to flooding that damages port facilities or may facilitate the transmission of some sea waves across the island's fringing reef to the shore where they could disturb operations in the harbour or damage it.

Rainfall and temperature. Nauru's climate is equatorial and maritime in nature. Annual rainfall averages over 2,000 millimetres (mm) per year. Variation is high, however, and periodic droughts are a serious problem. Heavy or extreme rainfalls can cause localized flooding, which can delay port operations. Temperatures are consistent throughout the year and close to the sea temperatures. Extreme temperatures may affect the working conditions of port staff and worker health.

Impacts of Current Climatic Conditions on Aiwo Port Facilities and Operations

Climatic conditions and the impacts of westerly wind and waves have played a part in the growing strain on the port's facilities and operations in recent years. Climatic factors exacerbate the weaknesses of the port's current mode of operations, and the port's weaknesses exacerbate the effects of climatic factors. The issues of climate and port design at Aiwo are intertwined.

A part-time port. Combined with the port's current shortcomings, the net result of the WPM and the swells from tropical cyclones has been to make Aiwo almost a seasonal operation. It has been closed for approximately 3 months of each of the recent years due to these climatic conditions and their effects on sea conditions.

⁸ Trenham et al., 2014. *PACCSAP wind-wave climate: high resolution wind-wave climate and projections of change in the Pacific region for coastal hazard assessments*. CAWCR Technical Report No. 068.

⁹ Collaboration for Australian Weather and Climate Research.

Facility damage. The damage to the port's still unrepaired mooring system occurred during an attempt by an oceangoing ship to manoeuvre away from the moorings during rough weather. The NPA advised that weather conditions were a factor in the incident.

Added operational costs. The costs of renting tugs from Solomon Islands to hold vessels in position for cargo transfer and handling in rough weather are high. When Aiwo cannot operate due to weather and the unloading and loading of cargo is attempted at the ill-suited, ill-equipped Anibare fishing harbour on the far side of the island, the process is slow and inefficient.

Diminished, imperilled sea link. The delays, technical challenges, added expenses, and risks caused by weather and the special difficulties of unprotected at-sea cargo handling at Aiwo port have affected the willingness of shipping companies to continue maritime transport runs to and from Nauru. Due in part to this, Matson Shipping suspended services to the country in November 2015, leaving Neptune Shipping the last company whose ships will ply the Nauru route. This has important consequences in terms of the country's shipping costs and the reliability of shipping services. It also raises the possibility that Nauru may soon not be served by commercial shipping at all.

Planned Infrastructure Improvements under the Baseline Scenario

The baseline scenario is a government plan to address the current infrastructure damage and operational challenges at Aiwo port through the following steps:

Fix and retain moorage system. Nauru would reinstall and maintain the currently damaged offshore mooring chain and buoy system. Some components that have reached the end of their service life would be upgraded or replaced. Once this step was completed, the at-sea cargo handling procedures would continue as before.

Employ tugs during rough weather. The rental and use of specialized tugs to assist these operations when the sea is rough would also continue.

Upgrade dilapidated shore facilities. New onshore facilities and buildings would be installed to replace dangerous and dilapidated existing ones.

Upgrade cargo handling equipment. New equipment for loading and unloading cargo and other port operations would be purchased and installed.

Current and future climate hazards unaddressed under baseline scenario. These baseline interventions would merely maintain the existing port design and operations and fail to address the growing risks posed to the port's functionality and viability by climate hazards.

Projected Climate Change and Impacts on Aiwo Port Operations and Facilities

This section looks at how climate change will alter the main climatic factors currently affecting Nauru's port operations and facilities. As discussed, these are sea level; westerly winds; westerly waves; and, potentially, rainfall and temperature. The section examines what impacts these changes and therefore climate change in general will have on the port operations and facilities under the baseline scenario discussed in the preceding section. Full details are provided in Annex 2N. Given the critical impact of westerly winds and waves on current port operations, this section looks at these factors first.

Wave Climate

The wave climate is currently contributing to the effective closure of Aiwo operations for about 3 months each year. The two forces directly driving westerly waves are westerly winds and distant tropical cyclones.

Westerly winds. Westerly winds prevail when the WPM prevails. The details of the factors determining the duration and intensity of the WPM are poorly understood. It is known that these are related to some extent to the El Niño Southern Oscillation (ENSO) and the location of South Pacific Convergence Zone (SPCZ). Confidence is very high that El Niño and La Niña events will continue to occur under climate change, but there is little consensus on whether climate change will cause El Niño and/or La Niña to change in intensity or frequency. The same uncertainty applies to the

impact climate change will have on the position and timing of the SPCZ. Climate change may or may not influence ENSO and/or SPCZ and so may or may not cause longer or stronger WPM. Hence, there is a possibility that climate change could lead to more frequent or more intense WPM.

Tropical cyclones. Tropical cyclones are a direct cause of high swell waves at Aiwo. The main factors determining the height of these waves at the port are the intensity of the cyclone and its distance from Nauru. Hence, should climate change affect the frequency, path, or intensity of tropical cyclones, this could have a major impact on swell waves at Nauru and the facilities and operations of Aiwo port. However, there is little consensus on how climate change will affect the path, intensity, and frequency of tropical cyclones in the South Pacific; some climate models project more frequent or intense tropical cyclones and some do not. This is another uncertainty factor. Climate change may not influence these cyclones, but it also may and therefore possibly cause more intense or more frequent swell waves at Nauru's Aiwo port.

Wave climate projections inconclusive. The IPCC notes that there is low confidence regarding projections for wind climate and wave climate. It does, however, note that 'there is increasing evidence for a strengthening wind stress field in the Southern Ocean since the early 1980s'.¹⁰ Further, the Collaboration for Australian Weather and Climate Research (CAWCR) has attempted to project how the wave climate at Nauru will change with climate change, taking the factors discussed above into consideration. The main CAWCR findings are below.

Wave change. The models predict only small or no changes in wave height, period, or direction.

Low confidence projections. For various reasons, the statistical significance is low. The results are presented with low confidence.

Tropical cyclone effects not considered. Importantly, the CAWCR models do not account for tropical cyclones, which are known to cause the largest waves. The findings state, "the predicted extreme values are underestimates due to not fully resolving tropical cyclones."

Consideration of other main factors unclear. It is not known to what extent the models account for possible shifts in ENSO, SPCZ, and WPM;

Findings inconclusive. The findings recommend that conclusions about the future wave climate at Nauru not be drawn at this time.

Little known about future wave climate at Nauru. These findings and other research so far show that little is yet known about whether climate change will affect tropical cyclone patterns, ENSO, SPCZ, and WPM and, in turn, about how climate change will affect wind waves and swell waves at Aiwo.

Wave Climate Risk Analysis. To address this uncertainty, more detailed analysis was applied to determine the levels of risk faced by Aiwo port. The findings are presented in Annex 2N and summarized below.

There is limited information available concerning downtime due to weather in port operations. The most complete data (reproduced as Table 4 below) comes from the Japan International Cooperation Agency (2014)¹¹ and indicates the number of days of operation lost through heavy weather.

Table 4: Number of days of port operation lost to heavy weather

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total days per year
2001	0	1	0	0	3	5	1	4	0	2	3	23	42
2002	8	10	0	3	2	4	14	8	19	20	8	18	114
2003	15	5	0	3	3	5	2	1	0	2	1	1	38
2004	21	3	7	5	0	7	4	8	0	0	0	11	66
2005	-	-	-	-	-	-	-	-	-	-	-	-	No data
2006	-	-	-	5	0	1	6	12	2	27	13	5	71 in 9 months

¹⁰ See Chapter 5 in IPCC (2013).

¹¹ Japan International Cooperation Agency, 2014. The Preparatory Study on the project of Constructing Reef-edge Quaywall and Causeway at Aiwo Harbour.

2007	-	5	4	0	2	0	0	0	0	0	4	5	20
2008	-	-	-	-	-	-	-	-	-	-	-	-	No data
2009	-	-	-	-	-	-	-	-	-	-	-	-	No data
2010	-	-	-	-	-	-	-	-	-	-	-	-	No data
2011	2	0	1	0	1	0	0	1	1	-	-	-	6 in 9 months
Total	46	24	12	16	11	22	27	34	22	51	29	63	

Source: Japan International Cooperation Agency, 2014.

Table 4 shows that 2002 was an exceptional year for downtime, particularly the second half of the year. It is noted from annual wind and wave roses (Annex 2N) that 2002 is a year of exceptional wave occurrence from the west and with larger waves than normal. The same analysis shows that the years 1997, 2002 and 2015 had stronger winds and waves from the west than other years. Using the CAWCR hindcast wave data, the operability for these years was compared with the average over all hindcast years. The results for the percentage of time with Hs above 1 m for varying durations are shown in Figure 8.

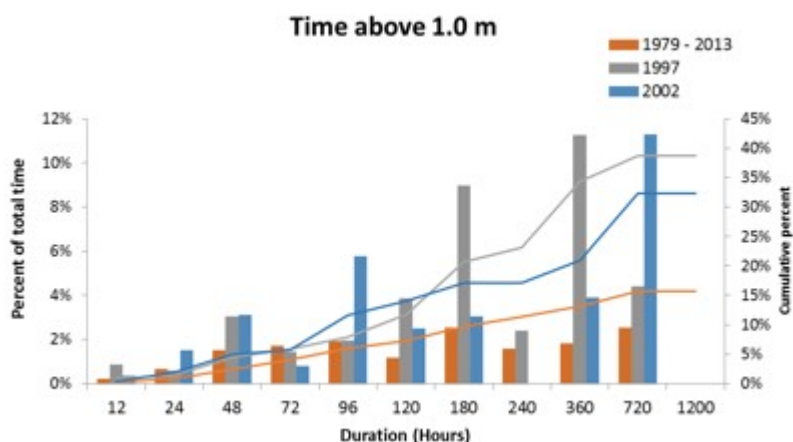


Figure 8: Port operability for a threshold of Hs = 1.0 m for varying durations

Figure 8 shows that both 1997 and 2002 have significantly more time with significant wave height above 1 m compared with the average, 39% and 32% respectively compared with 16% on average. It can also be seen that these exceedances occurred in longer durations with multiple periods between 240 and 360 hours (10–15 days) in 1997 and between 360 and 720 hours (15–30 days) in 2002. These are significant periods of interruption to the supply chain of essential supplies for Nauru.

The climatic characteristics of the three identified years were examined in more detail and each was found to be associated with a strong El Niño event¹². In each case this was associated with Madden-Julian Oscillations (MJO) which are atmospheric phenomena resulting in weakening of the Pacific trade winds, strengthening of the westerly winds in the WPM and westerly migration of the WPM with increased convective activity and rainfall. Modelling of the MJO in a climate change model has shown that a warmer climate supports higher amplitude MJO events, although with a reduced frequency.¹³ The MJO also tends to propagate further into the central and eastern Pacific in a warmer world.

It is of interest to note that the three years studied above, 1997, 2002 and 2015 stand out in the 35-year data set as having markedly greater frequency of westerly wind and waves and with higher wave heights. Whilst there is insufficient data to make a definite statistical statement, all three events occur in the second half of the data set suggesting there may be a component of climate change. This may also be coincidence, and a more detailed analysis

¹² Climate Assessment for 1997, Supplement to Bulletin of the American Meteorological Society, Vol 79, May 1998. State of the Climate in 2002, Supplement to Bulletin of the American Meteorological Society, June 2003. State of the Climate in 2015, Supplement to Bulletin of the American Meteorological Society, August 2016.

¹³ Subramanian, et al., 2014. *The MJO and global warming: a study in CCSM4*. Climate Dynamics.

would be required to determine whether this is the result of climate change or not, however it is consistent with such a trend.

As can be seen, there is a great deal of uncertainty regarding the future wave climate. There are some scenarios under which the wave climate becomes more favourable for shipping operations at Aiwo, but there are many under which the wave climate at Nauru worsens and will pose a growing threat to these operations under the baseline port improvement scenario. In these cases, the port operations would risk growing delays, danger, and cargo damage even with the baseline improvements and become increasingly expensive to undertake. Port closures could realistically increase by 100% over the average in some years. It is important to note that it is not the overall level of closures which is important but the increased possibility of longer durations of continuous port closure which is the major threat to the Nauru supply chain. While the port is dependent on slow at-sea transfer with each vessel taking 2—3 weeks to unload, the island is extremely vulnerable to long weather-induced delays.

This analysis above is broadly consistent with projections for increases in storm events. Under the B1 (low) emissions scenario, the MJO is also associated with increases in extreme rainfall events and these have been predicted to impact on Nauru.¹⁴ It is likely that the cause of this increase is the MJO, although this is not stated in the published report. Most models project that the current 1-in-20-year extreme rainfall events will average 3–4 times per 20-year period by 2055.¹⁵ This infers that something that now has a 5% (1-in-20 year) probability of occurring in any 1 year would have a 20% (4-in-20 years) probability of occurring in any 1 year by 2055. This is equivalent to a 1-in-5-year probability. If we assume that this same frequency increase affects wave events requiring port closure due to the unchanged at-sea cargo-handling process to be maintained under the baseline scenario, a 15% increase in the occurrence of port closure by 2055 can be expected. This should be considered a conservative estimate as it does not account for non-normal distribution of port closure, frequency and duration of closure events.

Sea Level

The available data indicates that the sea level rose near Nauru by about 5.0 mm a year between 1993—2016, i.e. more than the global annual average of 2.8 mm—3.6 mm. The reason for this increase is not known, but in many cases it is due to thermal expansion being greater in the tropics than in other areas. The mean sea level around Nauru is projected with high confidence to continue to rise over the course of the 21st century.

Table 5: Sea level rise projections in centimetres showing 5%–95% confidence interval in brackets

Emissions Scenario	2030 (cm/confidence interval)	2050 (cm/confidence interval)	2070 (cm/confidence interval)	2100 (cm/confidence interval)
RCP2.6 ¹⁶	12 (8%–17%)	22 (14%–30%)	32 (19%–45%)	42 (24%–60%)
RCP4.5	12 (7%–17%)	22 (14%–31%)	35 (22%–48%)	48 (29%–68%)
RCP6.0	12 (7%–16%)	22 (14%–30%)	34 (21%–48%)	49 (30%–69%)
RCP8.5	13 (8%–18%)	25 (17%–34%)	42 (28%–58%)	63 (41%–89%)

Source: Australia Bureau of Meteorology and CSIRO, 2014.¹⁷

From Table 5, to be precautionary, if the upper end of the 8.5 representative concentrated pathway (RCP) figure is adopted, port facilities should take precautionary steps to allow for a 0.34 m rise in sea level by 2050 and a 0.58 m sea rise by 2070.

The degree of sea level rise will have the following impacts on Aiwo port under the baseline scenario.

¹⁴ Australian Bureau of Meteorology and CSIRO, 2011. *Climate Change in the Pacific: Scientific Assessment and New Research - Volume 2: Country Reports*.

¹⁵ Australian Bureau of Meteorology and CSIRO, 2011. *Climate Change in the Pacific: Scientific Assessment and New Research - Volume 2: Country Reports*.

¹⁶ Represented concentrated pathway

¹⁷ Australian Bureau of Meteorology and CSIRO, 2014. *Climate Variability, Extremes and Change in the Western Tropical Pacific: New Science and Updated Country Reports 2014*.

Port shore facility flooding. Sea level rise will lead to overtopping of the harbour shoreline and more frequent inundation of port land and/or buildings.

Reef protection nullification. The protection provided by the reef to the small boat harbour may be reduced as sea levels rise and waves are more able to overtop it.

Temperatures

Temperatures in Nauru have increased and will continue to rise. This will mean more very hot days and warm nights and a decline in cooler weather. Under the high emission scenario, average temperatures may increase by 0.5°C–1.2°C by 2030. Extreme temperatures may affect the working conditions of port staff and worker health.

Rainfall

There is a high level of confidence in all climate change scenarios that the frequency and intensity of extreme rainfall events will increase. For example, most models project that the current 20-year average recurrence interval (ARI) daily rainfall event will become a 10-year ARI event for RCP 2.6 and a 5-year ARI for a RCP 8.5 by 2090 (BoM & CSIRO, 2014). This may cause localised flooding at the onshore port facilities and contribute to greater delays, danger, and operating costs.

Conclusion

The CRVA (refer Annex 2M) includes an assessment of climate risk to the existing port from current and future climate. It clearly shows that the existing port is not climate resilient both in the present day, and under climate change conditions. It was therefore necessary for the strategic options assessment for the project to consider a new port.

Table 6 summarizes the current climate hazards affecting the port, how they may evolve with climate change, and the impact this will potentially have on the port operations and facilities under the baseline improvement scenario. (source: CRVA, Annex 2M).

Table 6: Summary of climate hazards and climate change risks

Hazard	Potential climate impact	Potential impact on port operations and facilities
Westerly wind driven waves	Risk of increased frequency and intensity	Offshore at sea, ships unable to moor, leading to increased costs and delays.
		Increased risk of damage to moor, moorings or fuel and phosphate transfer systems – leading to increased costs and downtime.
		At sea ships unable to moor – leading to increased costs and delays.
Cyclone-driven swell waves	Risk of increased frequency and intensity	Loading and /unloading offshore at sea is less manageable – due to difficulty holding ships in position – leading to greater delays, damage, and danger.
		Offshore at sea, ships unable to moor, leading to increased costs and delays.
		Increased risk of damage to moorings or fuel and phosphate transfer systems – leading to increased costs and downtime.
Waves/mean sea level	Sea level rise, up to 58 cm by 2090	Offshore loading and unloading at sea is potentially less manageable— due to difficulty holding ships in position – leading to greater delays, damage, and danger.
		Increased risk of overtopping of harbour wall and damage to facilities. Rebuilding may be necessary.
Temperature/heat wave	Increased temperatures and temperature extremes	Risk the current system of offshore buoys and moorings will need replacing for new sea depths.
		Higher temperatures for workforce may slow down freight handling times.

Extreme rainfall events	Increased and more frequent extreme rainfall	Although not a problem in the past, flooding of harbour land and/or buildings may occur in the future.
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Key climate change risk factors. The key climate change impact on Aiwo port is the risk that increased intensity of those events (e.g. ENSO, MJO, TCs) that generate westerly winds and waves will make operating lead to increases in the port under the baseline scenario frequency/duration of downtime of make the less feasible. The port may be closed more frequently. The port operations may become costlier as the adverse weather and wave conditions become more common and require more frequent expensive rentals of special tugs to secure ships during the at-sea cargo handling processes. A worsening of delays and difficulties at the port may lead to demands by the current shipping operator for higher fees, or the operator may stop serving Nauru altogether.

Potential for steady degradation of port operations. There will not be one point in time when adverse wave conditions bring Aiwo port's operations to an abrupt complete stop and effectively cut Nauru's critical sea link with the world. The risk rather is that such wave changes in combination with sea level rise and other non-climatic factors inherent in maintaining the current operational mode under the baseline scenario will make the port less reliable and less effective and more costly and dangerous. The port operations would gradually become unfeasible first as a commercial venture and then in any form. Long before then, moreover, the port will have ceased to provide the reliable, efficient maritime lifeline Nauru must have to sustain its economy, its socioeconomic development, its culture, and the livelihoods and well being of its people.

Precautionary principle. There is some - but not full - scientific certainty that climate change will render the baseline project (i.e. fixing the existing buoy system) inoperable within the next 50 years. Notwithstanding the lack of full scientific certainty, the Government of Nauru does not want to postpone its decision to reject the baseline project and develop a port system, which will certainly be climate resilient under all future climate scenarios. In short, the Government has made a strategic decision to invoke the UNFCCC's Precautionary Principle (Article 3.3): "The Parties should take precautionary measures to anticipate, prevent or minimize the causes of climate change and mitigate its adverse effects. Where there are threats of serious or irreversible damage, lack of full scientific certainty should not be used as a reason for postponing such measures". In this case, applying the principle is a fair and 'no regrets' course of climate adaptation action for Nauru, and appropriate given the importance of the port to all aspects of life on Nauru and the serious harm caused by climate change that could occur if action is delayed. It is also important to note that Nauru (as a Party to the UNFCCC) and the GCF (whose Governing Instrument is guided by the principles and provisions of the Convention) recognize the importance and validity of the Precautionary Principle. Therefore, all parties should give equal value to the Precautionary Principle and its application to this proposal.

The Proposed Alternative

To manage this risk, the alternative proposed to the baseline scenario is the construction of a port at Aiwo that is protected from winds and waves. This protected port should be capable of servicing oceangoing ships and providing a relatively calm environment for efficient and predictable loading and unloading all year-round. Port closures should be rare. Given the limited projected effects of climate change at Aiwo, the port should be (i) protected from westerly winds and westerly swells, (ii) adapted to sea level rise, and (iii) adapted to potential temperature rises and increases in extreme rainfall events.

Such a port would give Nauru the efficient, reliable, climate-resilient facilities for all maritime transport operations it needs to deal with its climate change future. It would have major immediate and long-term benefits for the country's economy, socioeconomic development, and security, as well as for the health, nutrition, education, and culture of its people.

Barriers to the Proposed Alternative

Nauru confronts three main obstacles to constructing and operating the proposed protected port, although an optimal design has been selected from five originally considered based on technical, economic, and climate change criteria and considerations. (The full feasibility study is in Annex 2.) The barriers to proceeding with the alternative rather than the baseline scenario are the following:

1. **High up-front investment costs.** The initial investment required to build the protected, year-round port is about \$65 million. Nauru cannot provide this amount of funding and does not currently have access to enough financing.
2. **Current operational context.** To operate the proposed protected, safe port reliably, effectively, and with minimal delays year-round, the port's governance and operations need to be reformed. This will also be necessary if more international shipping companies are to be attracted to servicing the port and thereby increasing competition on the Nauru route. Improvements are needed in the areas of ownership, overall management, financial management, fees, storage facilities, processing facilities, loading equipment, and customs services.
3. **Human resource capacity.** Management, maintenance, and operation of the proposed port will require well-trained personnel to fill many roles. Many of the port's staff members have little training, and there currently are no suitable candidates for many of the positions required in a modern port.

Critically, if GCF finance is added to the project it would remove the first major barrier to proceeding with the proposed climate resilient alternative – the high up-front investment costs.

Climate Change Mitigation

Without project Container ships and fuel ships spend many days at sea whilst unloading. Climatic conditions mean they have to use the engines almost continuously during this period in order to 'stand still'. This leads to significant fuel use and CO2 emissions. In addition, the pusher barges make many trips to/from the container ships to assist loading/unloading, with further fuel use and CO2 emissions.

With Project Container and fuel ships will have a short wait before traveling to wharf and docking. Their fuel use will be greatly reduced. Likewise, use of pusher barges will be reduced to the simple operation of helping the container and fuel ships into the wharf – achieving additional reductions in fuel use.

The net result is a significant reduction in fuel use and GHG emissions.

C.3. Project / Programme Description

Goal, Objective, Components, and Outputs

The project goal is to ensure Nauru's regional and international economic, cultural, and social connectivity is sustainable and climate-resilient.

The project objective is to establish a climate-resilient port with efficient and sustainable operations that provides the basis for regional and international maritime transport. To achieve this, the project has two components, with one output under each (therefore the words 'component' and 'output' are interchangeable when discussing the proposed project).

Component 1: A protected port

Output 1: Priority port infrastructure constructed or rehabilitated and climate proofed

Component 2: Port reforms

Output 2: Reforms for the NPA implemented

Outputs and Activities

Output 1: Priority port infrastructure constructed or rehabilitated and climate proofed

Delivery of this output will make Aiwo's port facilities accessible and usable by oceangoing ships all-year round. It will make the port resilient to climate change. Ships will travel directly to Nauru's coastline where they will be able to manoeuvre, dock, and be loaded and unloaded directly from and to the shore in calm conditions and without the need

for trouble-plagued at-sea transfers of cargo between ships and barges. This will make the port's cargo operations more efficient and predictable and eliminate the current constant delays and interruptions. The new facilities will also make Aiwo a more attractive port of call for shipping companies and should be an incentive for second and possibly more shipping companies to begin service to Nauru. The overall results of this output's delivery will be greatly lowered shipping costs, much fewer delays, and reduced damage and danger during the port's cargo operations. Among the major advances that a protected, climate-reliant new port will provide are the following:

- Minimize downtime, due to protection from most adverse weather and wave conditions and the end of off-shore cargo handling.
- Elimination of dependence on the time-consuming, frequently delayed, and dangerous transfers of cargo at sea.
- End of the port's reliance on the aging and damage- and breakdown-prone mooring buoy system and cantilever loader infrastructure.
- Reduction in freight costs, resulting from faster turn-around times and lower demurrage for container and general cargo vessels.
- Safer and quicker transfers of Nauru's fuel imports from tankers on to the island.

Six options were considered, including the baseline scenario. The options were assessed against the following criteria:

- Technical feasibility—see feasibility study report and concept drawings (Annexes 2 and 2A).
- Economic feasibility—see cost estimates, economic feasibility assessment, and financial analysis (Annexes 2C, 2D and 2F).
- Operational feasibility—including model simulations of ship docking under diverse conditions (see Annex 2B).
- Resilience to meteorological, climatic and ocean hazards—based on the findings of the Metocean study (Annex 2N) and the CRVA (Annex 2M).

The option selected consists of a channel through the island's surrounding reef between the ocean and the coastline, a wharf for docking ships, a dredged berth for turning and docking, and a breakwater wall to protect the ships from waves when docking and loading or unloading. The general layout is shown in Figure 8.



Figure 8: The design for the proposed climate-resilient port

Delivering this output will involve the following activities:

1. Implementing and completing civil works:
 - Construction of a wharf structure 140 m long and 30 m wide, and of an adjacent 50 m wide berth pocket dredged into the reef to a depth of 10 m.
 - Construction of a 170 m long breakwater on the north-western side of the wharf area to provide protection from waves.
 - Dredging of a navigable canal between the ocean and the wharf through the reef.
 - Construction of one storage, one office and one terminal building and internal roads and parking for trucks, cars, and bikes.
 - Construction of a yard for containers.
 - Construction of perimeter fencing and other security installations.
 - Provision of port building services, such as drainage, water mains, electricity, a security building, and an X-ray and weighbridge building.

Full details of the selected option, the justification, and the design details are provided in the annexes, notably Annex 2 (main report) and Annex 2A (concept drawings).

Output 2: Reforms for the NPA implemented

The project will undertake several activities to help make the governance and management at the new Aiwo port effective. This will include the following:

1. Recruit port reform consultants
2. Approve (by Nauru Parliament) SOE and Tariff Reform Legislation

3. Approve Port Regulations (by Nauru Cabinet and Minister of Ports)
4. Agree on new Organisation Structure
5. Prepare and establish financial management system and processes
6. Prepare and establish asset management system and Processes
7. Prepare and implement operation and maintenance plan
8. Prepare and implement Human Resource Management and Training Plan
9. Recruit necessary staff for NPA
10. Recruit Chief Operation Officer (International) for NPA
11. Procure port equipment, tools and software
12. Reform implementation and management

The State Own Enterprise (SOE) model is recommended as the appropriate management model for NPA. This recommendation was concluded upon detailed assessment of other potential options including contracting port management and operations to a specialist private sector company and possible public private partnership (PPP). The primary issue is the ability of the NPA to generate sufficient revenue for the operations and lack of necessary skills of the NPA workforce in operating port activities efficiently. The effect of this is it is unlikely that any contracting or PPP would work effectively. However, the proposed reforms and new port facilities will attract new business opportunities for NPA and the PPP option can be considered consequently possibly after 3-5 years upon establishing new port facilities. The project will support NPA to implement reforms for at least 5 years.

The reforms and capacity building of NPA are contingent for successful port management, operations and sustainability of the new assets acquired under the project. Under ongoing project preparatory works, a detailed study is being undertaken to assess actual gap in the NPA's overall organization management and operations to function as a credible organization complying to international best practices. The assessment concluded that (i) NPA need to be restructured as a State-Owned Enterprise (SOE) against current government's department type management, (ii) establish and implement a cost recovery level tariff structure and regulations, (iii) establish new IT based accounting, financial, human resource and training, operation and asset management system and procedures, and (iii) capacity building of port authority staff through on-the job and specific training. The government has agreed to institute necessary legislative approvals for transforming NPA into a SOE, revision to tariff structure, and introduce or revise port related regulations. It is planned that legislative and regularity amendments and new organization structure, procedures and practices can be established by mid-2018. By the time the new port assets will be completed by end 2019, the NPA will be expected to substantially functioning as a credible organization compliant to international standards especially the requirements of international maritime organization (ISPS Code) and UN maritime conventions (UNCLOS) which government is signatory to. ADB will manage and oversee overall implementation of port reforms and capacity building initiatives until end of 2022 with possible extension depending on the requirements to be assessed at that time. In support of this ADB is currently arranging recruiting and appointing an international consultant for NPA's chief executive officer position (effective 1 July 2017 until end of 2022) and other key long term management positions subsequently.

C.4. Background Information on Project / Programme Sponsor (Executing Entity)

The project sponsor is the Ministry of Finance (MOF). The project executing entity is the Planning and Aid Division (PAD) of MOF. A project steering committee has already been established to support MOF. The project implementing agency will be the NPA. A project management unit (PMU) will be established to implement the project within the implementing agency.

A summary of the responsibilities of the executing and implementing agencies is provided in Table 7.

Table 7: Roles of key project agencies

Agency	Roles
Accredited Entity: ADB	- Oversees the project (preparation and implementation) - Administers ADB's grant to the Government of Nauru - Administers the co-finance resources from the Government of Australia in accordance with the co-financing agreement - Administers the co-finance resources from the GCF in accordance with the co-financing agreement - Implements Monitoring and Reporting procedure agreed with the GCF Secretariat

	• Submits annual performance report to GCF • Monitors project implementation arrangements, disbursement, procurement, consultant selection, and reporting • Monitors schedules of activities, including funds flow • Reviews compliance with agreed procurement procedures • Reviews compliance with grant covenants • Monitors effectiveness of safeguard procedures • Analyzes the outcome of the capacity building and training programs • Monitors conformity with ADB anti-corruption policies • Undertakes a periodic review mission • Undertakes a joint midterm review mission • Undertakes project completion report • Independent Evaluation Department validates project completion report and has option to conduct an evaluation
Executing Entity: MOF	• Represents the government as a grant recipient • Facilitates negotiation, signing, and execution of the project financing agreement • Reviews and approves the project's procurement actions, in coordination with MOT • Submits withdrawal applications to ADB • Expedites implementation and minimizes cost by ensuring necessary counterpart funds are available on time; reviewing invoices and payments to contractors, consultants, and other service providers; and monitoring project progress and instructing MOT to take corrective actions to prevent significant variations and deviations from schedules and budgets
Project Steering Committee	• Facilitates the exercise of responsibilities by the executing entity • Oversees project implementation • Monitor project progress • Resolves any obstacles or impediments to implementation • Guides the executing entity Project Steering Committee consist of (i) secretary, Ministry of Finance (chair); (ii) deputy secretary for planning and aid, Ministry of Finance; (iii) advisor to the Ministry of Finance; (iv) chief executive officer, NPA; (v) ADB; (vi) the Government of Australia, (vii) other donor partners.
Implementing Agency: the NPA	• Coordinates with ADB for all project management activities • With the support of the PMU, manages and monitors project implementation activities, ensuring compliance with the government and ADB's requirements • Initiates Task Force meetings as necessary and at least quarterly • Reviews monthly progress reports that PMU prepares and endorse to the task force • Supervises PMU in providing government counterpart assistance to Supervising Consultant (PSC) • With the support of PMU, coordinates with and supervises PSC • Facilitates the coordination with government agencies necessary to prepare and implement the project • Oversees reporting and monitoring of project performance, including preparation of monthly and quarterly project progress reports • Oversees and approves for recommendation to higher authorities' contract administration matters, in close coordination with PMU • Assists ADB during project review missions

A full assessment has been undertaken of the financial management and procurement capacity of the executing and implementing agencies (Annex 2E).

The financial management assessment concluded that the executing entity (MOF) presents a low risk due to its experience in managing and implementing projects financed by ADB and other international funding institutions.

However, the assessment determined that the risks associated with the implementing agency (NPA) to be high due to its lack of experience in managing and implementing projects funded by foreign institutions, including ADB-funded projects. To address these risks, the assessment recommended 14 risk mitigation measures. Some of these are already under implementation supported by ongoing project preparatory technical assistance and project design

advance include (i) disbursement of funds directly by ADB, (ii) consultant selection fully delegated to ADB and recruitment of consultants through ADB CMS, (iii) overall management and consultant performance supervised by ADB in consultation with NPA and the Ministry of Finance, (iv) periodic review missions by ADB and the donor partners, (v) enhancing government ownership of the project by fully engaging Ministry of Finance, NPA and other stakeholders in project preparatory and detailed designs of the project components, (vi) agreement by the government to engage an international consultants for NPA's Chief Executive Officer position enabling effective and efficient implementation of the project and the proposed NPA reforms. Other mitigation measures that will be supported by the project during implementation will include (i) providing technical assistance (TA) for the institutional strengthening of the NPA, for development and implementation of a financial management information system and accounting procedures complying to generally accepted accounting principles (GAAP) and standards and for an asset management and maintenance plan; (ii) providing TA to undertake a tariff study and reforms for the ports; (iii) including a financial management specialist in the consultancy teams to support project accounting and financial reporting; and (iv) providing training in ADB procedures related to procurement, disbursement, and project management.

In addition, the following financial covenants are being incorporated into the grant agreement for this project:

- The NPA shall prepare regular financial reports and shall be audited on an annual basis by an external auditor acceptable to ADB complying internationally accepted standards for financial reporting and auditing.
- ADB shall disclose the annual audited financial statements for the Project, and the opinion of the auditors on the financial statements, within 30 days of the date of their receipt by posting them on ADB's website.
- The NPA shall install computerized financial management systems based on the new port operations, in line with international best practice.
- The NPA shall hire qualified accounting and finance staff to perform the necessary financial management function.
- Financial performance indicators shall be reported on a regular basis, including profitability ratios.

The conditions for withdrawal applications include an invoice by the contractor/s, explicit recommendation by supervision consultants that work done by the project contractor/s are duly satisfied to the contract specifications, and payment certificates are duly signed by the supervision consultant and approved by the implementing and executive agency authorized signatories. All project related expenses will be disbursed following guidelines as set out in ADB's Loan Disbursement Handbook, 2015.

Finally, the ADB, through its Pacific Liaison and Coordination Office in Australia, will backstop, monitor and periodically review the project's progress.

C.5. Market Overview (if applicable)

The principal impact of this project is to create a climate-resilient port for Nauru. Nauru's population is about 11,000, which means that only one protected international port is justified for the island. The port can be treated as a public good to be owned by the state and managed under state guidance and not for individual profit. The port can be treated as a monopoly, since it will be the only entity providing international port facilities.

Nauru is currently one of the most expensive (and possibly the most expensive) port in the region for shippers and shipping companies to use. This is due largely to its inefficient current design and mode of operations. Nauru may be at risk of losing the service of Neptune Pacific shipping line (Annex 2D). After the withdrawal of services by Matson Shipping in November 2015, Neptune Pacific became the sole transporter of freight to Nauru. This provided the shipping line to increase freight costs by a significant amount, which has in turn greatly affected the prices for consumers in Nauru. The new port will make port operations more attractive to shipping lines and facilitate competition. This should translate into improvements for users and importers.

This notwithstanding, the port operations and management are to be subjected to the highest forms of supervision to ensure efficiency and accountability. The port reform actions under component 2 will ensure this. Notably, a strong tariff reform element will ensure the following:

- Shipping companies will pay appropriate and equitable tariffs.

- Shipping costs will remain efficient and competitive.
- Appropriate revenue will accrue to the NPA and on to the government.
- There will be no undue delays in the provision of services or the collection of revenue.

The financial analysis (Annex 2F) provides projections for revenue and expenditures for the NPA until 2027. No government subsidies are expected.

A small, less protected fishing harbour exists at Anibare on the east coast of Nauru. This harbour does not provide any sizeable permanent loading, storage, or processing facilities. It does not provide a similar service to the proposed port at Aiwo and so is not a competitor. Operations at Anibare should not be affected by the upgrade to Aiwo port.

One major indirect impact of the upgraded Aiwo port will be greatly improved access to regional and international markets for stakeholders on Nauru. Individuals and enterprises on Aiwo will be able to interact more efficiently with regional and international markets and become more competitive on the regional and international stage. It is expected that this will promote economic activities on Aiwo. For example, the project will upgrade fish storage and processing facilities on Aiwo, enabling Nauru to play a more important role in this sector.

C.6. Regulation, Taxation and Insurance (if applicable)

Taxation Regulation

ADB projects in Nauru are exempt from taxes, including import tax. There are no regulations affecting the import or use of foreign currency under this project.

Permits

Land use and building permits will be required in line with Nauru regulations for the construction and operation of the port and port facilities. The concerned government agencies are MOF, MOT and the NPA. They have the authority to obtain all necessary permits and licences (MOF, MOT), and manage and operate the port (NPA). The budget includes all costs expected for obtaining permits, notably for obtaining access to drilling sockets.

Insurance

The budget includes all insurance costs associated with contractors and the construction. This includes (i) professional indemnity insurance for design and supervision consultants covering detailed engineering designs and construction supervision, (ii) insurance for contractors works and equipment (All Risks Insurance Policy) and (iii) third party insurance covering injury to persons and damage to property outside of the contractor's responsibility.

Legislation

The project will comply with all national laws and regulations, including those listed below.

- Port Authority (Amended) Act 2016 and Regulations 2016
- Disaster Risk Management Act 2008
- Marine Pollution Prevention Bill (drafted in 2000)
- Litter Prohibition Act 1983
- Wild Birds Preservation Ordinance 1937
- Land Act 1976
- Explosives Ordinance 1924

C.7. Institutional / Implementation Arrangements

The project management arrangements are summarized in the diagram in Figure 9 and the financial flows are illustrated in the diagram in Figure 10.

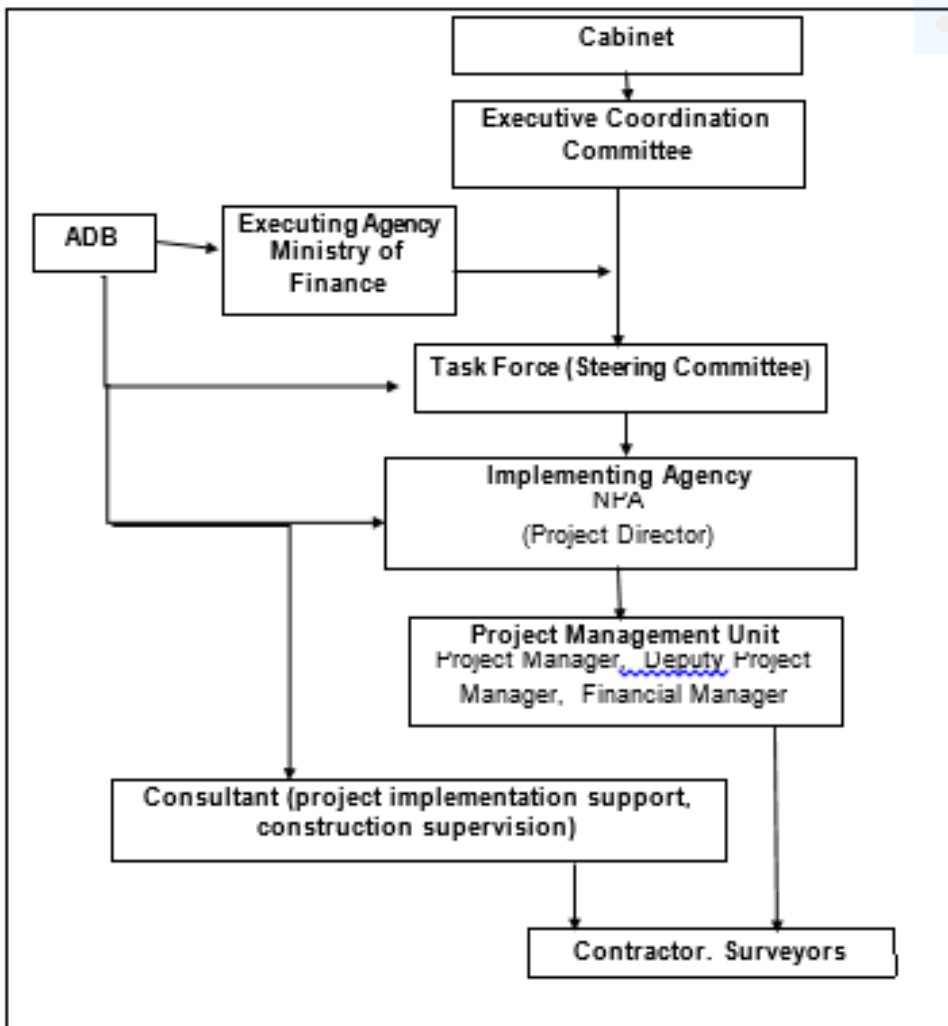


Figure 9: Project management arrangements

Note: Refer Table 7 for role and responsibility of Ministry of Finance as the executing entity and NPA as the implementing agency.

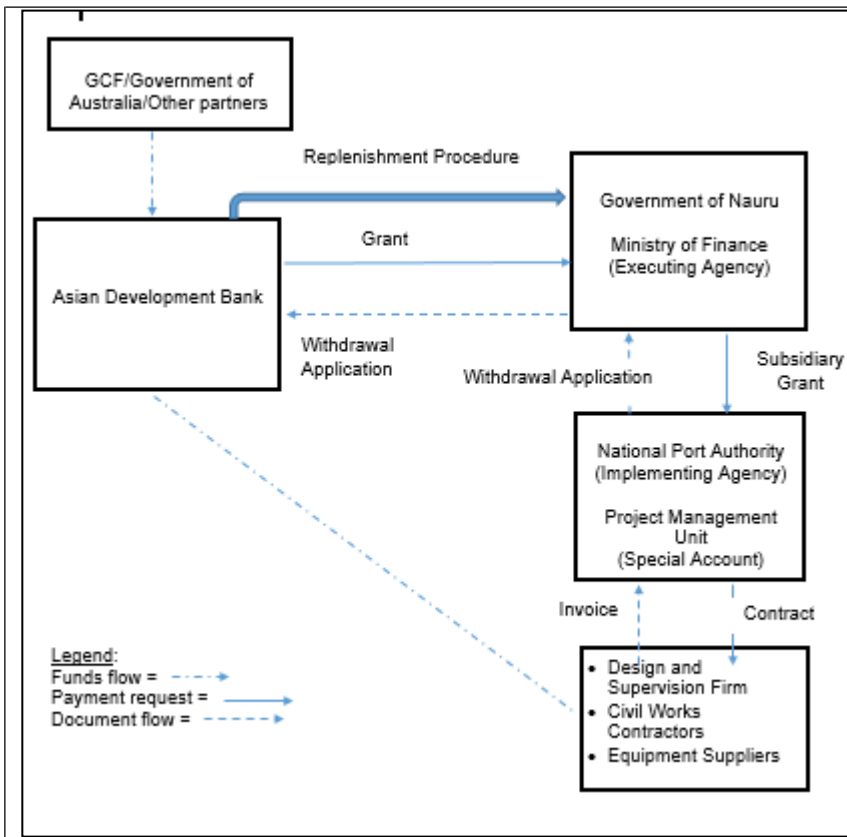


Figure 10: Financial flows

Figure 10 shows there are two cofinancers: Australia and GCF. Both provide their funds to the financier, ADB. ADB administers all cofinancing funds in accordance with the Grant Agreement with Nauru, which will be consistent with ADB's Operations Manual. The Government of Nauru provides counterpart funds. They are managed internally within Nauru. The amount of funds and what they will spend it on will be in ADB's Report and Recommendations to the President and the Project Administration Manual.

Construction and Supervision Methodology

Construction of the port facilities will be carried out by a construction company selected through an international competitive bidding following ADB's Procurement Guidelines (2015, as amended from time to time). The contractor will be directly supervised by an external construction supervision consultant (a firm) with experience in marine construction recruited through competitive bidding following ADB's Guidelines on the Use of Consultants (2013, as amended from time to time). ADB will assist NPA for selecting a contractor and construction supervision consultant firm.

Post Project Completion Arrangements

When the Project is completed, the infrastructure will be handed over to the Government of Nauru, per standard contract procedures. The project will help strengthen the mechanisms for monitoring, operations and maintenance (O&M), and building associated institutional and community capacities for executing the O&M during and beyond the project's implementation. This will help ensure that the infrastructure is managed sustainably after the turnover.

C.8. Timetable of Project/Programme Implementation



D.1. Value Added for GCF Involvement

Under the baseline scenario, the port facilities and operations will continue to deteriorate and be threatened by climate change. This is not a climate-resilient scenario. There is a credible risk that under the baseline scenario the port will reach a tipping point where due to climate change it is effectively inoperable. The estimated cost of improvements and upgrades to baseline scenario is \$28.4 million.

Nauru is a SIDS and is highly vulnerable to external economic shock and to climate change. Its small size, remoteness from partners and markets, and limited economic diversification opportunities mean it cannot access commercial credit on favourable terms. As a SIDS with high debt levels and as a nation highly vulnerable to climate change, Nauru requests grant financing.

GCF financing will add value by supporting the construction of a climate-resilient alternative to the baseline scenario that will leave the port open to the real risk of future operational degradation and breakdown due to climate change. The alternative will deliver real lasting benefits to the population of Nauru and secure the maritime transport link and sea connectivity it is totally dependent upon.

The GCF-supported climate resilient alternative has relatively high up-front costs of \$65.20¹⁸ million. This is \$36.80 million more than the baseline cost. Although the baseline will deliver benefits through some upgrades by restoring the port's previous climate-vulnerable mode of operations, the port cannot become climate-resilient without this additional investment. For this reason, the \$36.80 million in additional investment represent climate change adaptation costs. The small island nation of Nauru is unable to cover these costs or to borrow this level of funding. GCF is therefore requested to contribute \$26.91 million to help finance the climate-resilient alternative to the baseline scenario project. The requested support of \$26.91 million will represent

- 41% of the overall \$65.20 million project costs, or
- 73% of the \$36.80 million of costs eligible for climate change adaptation funding.

The other financing partners are the Asian Development Bank, the Government of Australia, and counterpart financing by the Government of Nauru. Together, these partners will ensure that Nauru and its people will have a climate-resilient port that can provide safe, reliable, efficient maritime transport services all year-round.

The GCF involvement will add value to the baseline scenario, which is not climate-resilient, by making the port climate resilient. Its support will help do this in three main ways:

Construction of channel to the coast, GCF contribution \$4.60 million (including physical contingency). The alternative project will dredge a channel for oceangoing ships through the island's fringing reef between the coast and the open sea. The baseline project includes no channel through the reef, which means that these ships and cargo transfer operations using barges will remain exposed offshore to the mounting weather and wave hazards due to climate change. Port operations will be delayed, dangerous, and intermittent, especially during the WPM. The alternative project will enable oceangoing vessels to come all the way to shore into an area protected from climatic hazards and climate change effects to load and unload directly to and from a wharf at the port's landing area.

Construction of a breakwater, GCF contribution of \$3.80 million (including physical contingency). The baseline scenario project will leave oceangoing ships and offshore loading and unloading operations in the open sea and thus permanently vulnerable to the current and a future worsening weather and wave climate. Only small boats and the barges that must now ferry all cargo to and from the shore and oceangoing ships can use the small 110-year-old protected harbour that currently exists (and can do so only in favourable weather). The breakwater to be constructed will protect the large oceangoing ships that will come directly to shore for loading and unloading from heavy weather and waves. It will create a calm sea environment that will be sheltered from future climate change effects and ensure safe, reliable, year-round cargo handling.

Construction of a new wharf, GCF contribution of \$18.51 million (including physical contingency). Providing a new wharf is the third essential part of the climate-resilience components of the alternative project, combining with the channel and breakwater construction to create an area for berthing oceangoing ships and loading and discharging cargo and passengers directly to and from the island that is protected from the effects of extreme weather and waves and climate change. The existing wharf is suitable only for the flat barges that currently must travel to and from the small protected harbour to ferry cargo and passengers between oceangoing ships left at sea and the port landing area. This process, vulnerable to the current and future delays, shutdowns, and potential accidents under climate change, would continue under the baseline project scenario without GCF's value addition and the alternative project.

Protection from sea level rise. The design of the new port under the project is adapted to sea level rise. Specific features to raise infrastructure heights have been incorporated into the breakwater, the wharf, and the foreshore construction to raise crest levels and make facilities more resistant to wave overtopping and damage under sea level rise conditions. The cost of these measures is \$4.20 million. This figure is included in the total GCF costs of \$26.91 (see details in the Climate Risk and Vulnerability Assessment (CRVA) in Annex 2M) and adapted to sea level rise. Specific features to raise infrastructure heights have been incorporated into the breakwater, the wharf, and the foreshore construction to make facilities more resistant to sea level rise.

D.2. Exit Strategy

The exit strategy is to ensure that the alternative project will lead to sustainable port operations and maintenance that continue *without* a need for support from international partners. The project includes several features to achieve this. Firstly, the port design and construction materials selected are chosen to be highly resilient and to have a minimum design life of 50 years. Secondly, the government's \$4.09 million contribution to the investment and \$6.27 million in tax and duty exemptions and in-kind contributions demonstrate its commitment.

Finally, the Government is fully committed to the port reform activities under component 2 of the project. These activities will strengthen sustainability:

Governance. The port reform activities will clarify the roles of the various stakeholders—notably MOF, MOT, and the NPA. The operation and management of NPA will be undertaken as a State Owned Enterprise (SOE) and the government will institute necessary legislation enforcing this. The government has already approved recruiting and engaging an experienced international expert for Chief Operation Officer (CEO) position of NPA for overall management and operation of the port authority following international best practices. In addition, the government will approve engaging international experts to other key positions of NPA including financial management and port operations. These experts will train NPA staff and gradually hand over responsibilities of the respective positions. This will create a basis for responsible decision making for all port investments and operations. These activities will clarify ownership of assets and of revenue. This will improve transparency and create a sound basis for income generation and partnerships.

Financial stability. Financial stability. The project will introduce port tariff reforms and take specific measures to improve revenue collection capacity. These measures will put the port on a firm financial footing. Annex 2F shows projections of the financial revenue and expenditures of the NPA. Revenue is projected to increase from AUD\$143,074 (US\$107,615) in FY2016 to AUD\$158,450 (US\$119,270) in FY2019. Expenditures are projected to decrease from AUD\$134,653 (US\$101,334) to AUD\$130,570 (US\$98,245) over the same period. This demonstrates that financial sustainability is feasible.

Operational stability. The reform activities will transform NPA's organisation structure under a SOE model and clarify responsibilities through clear job descriptions. The reforms will introduce a comprehensive capacity development program for NPA and provide necessary training for individual divisions and staff to ensure that the NPA can operate the port facilities at modern standards. This will cover financial management, asset management, and operational issues.

¹⁸ Excludes in-kind contribution by Government of Nauru, valued at \$6.27 million

Maintenance. The project will develop a financially backed operations and maintenance plan. The costs for this are included in the financial assessment (Annex 2F). The tariff and other reforms will ensure that the finance required for maintenance is available.

Capacity building. A full training needs assessment is being undertaken and will be followed by a broad training programme. This will give all stakeholders the capabilities required to operate the port and provide port-related services. This is an activity under component 2 (Port reforms).

E.1. Impact Potential

Potential of the project/programme to contribute to the achievement of the Fund's objectives and result areas

E.1.1. Mitigation / adaptation impact potential

Mitigation Potential

In both with and without project scenarios, all ships use equal time and fuel to reach Nauru waters. However, in the 'with project' scenario, the new port facilities will mean that ships spend considerably less time at sea whilst being unloaded/loaded, thereby leading to significant reduction in CO₂ emissions from the shipping sector. The climate mitigation impact would be achieved annually, and every year the port operates it would increase its impact potential.

Without project. Container ships and fuel ships spend many days at sea whilst unloading. Climatic conditions mean they must use the engines almost continuously during this period in order to 'stand still'. This leads to significant fuel use and CO₂ emissions. In addition, the pusher barges make many trips to/from the container ships to assist loading/unloading, with further fuel use and CO₂ emissions. Annually, during this loading/unloading process, 4630.2 tons of heavy fuel are used by ships and barges, and GHG emissions also occurs due to the need for electricity consumption at the shore facilities. It has been estimated that 27,529 liters of diesel will be consumed to produce electricity needs in the baseline facilities. The estimated baseline emissions representing these emission sources is 14,492 tCO₂e per year.

With Project. Container and fuel ships will have a short wait before traveling to wharf and docking. Their fuel use will be greatly reduced. Likewise, use of pusher barges will be greatly reduced to the simple operation of helping the container and fuel ships into the wharf – achieving additional reductions in fuel use. Annually, during this loading/unloading process, 851.2 tons of heavy fuel are used by ships and barges. The new port facilities will also require electricity. It has been estimated that about 422,885 liters of diesel will be consumed to generate the electricity needed at the new port facilities per year. The estimated GHG emissions from all the project emissions sources listed above is 3,784 tCO₂e per year. The GHG emission reduction impact of this project has therefore been estimated to be about 10,708 tCO₂e per annum. Details of the emissions calculations are presented in Section E.1.2.

Adaptation Potential

The project will help 100% of the population of Nauru (about 11,300) adapt to climate change. The port provides a transport service directly affecting the lives and livelihoods of the entire population.

Without a climate resilient port, Nauru will experience increasing food and health insecurity, energy insecurity, and water insecurity due to the failing port operations and limited connectivity. It will also experience progressively higher transport costs and increasingly unreliable transport facilities that will undermine any chance of new economic developments. Due to climate change, Nauru will become more and more disconnected from regional and international partners, markets, development, trends, and knowledge.

By providing climate proof maritime transport facilities, the project will:

- Provide increased food security and water security to the islands population;
- Provide increased access to health and education goods and services to the island's population;
- Provide increased economic opportunities, and ultimately jobs and government revenue (specifically, the project will establish processing and storage facilities at the port (not financed by GCF) to facilitate fishery sector development).

All infrastructure provided under the project will be built to the highest standards and to achieve climate-resilience. All infrastructure will have a minimum 50-year design life under the most extreme climate change scenarios. Therefore, the impact on resilience will be immediate because the design is for a climate-resilient port over its design life. This adaptation and improved resilience is essentially 'locked-in' from the first day of the new ports' operation.

E.1.2. Key impact potential indicator

Provide specific numerical values for the indicators below.

GCF core indicators	Expected tonnes of carbon dioxide equivalent (t CO ₂ eq) to be reduced or avoided (Mitigation only)	Annual	10,708
		Lifetime	535,400 (port lifetime of 50 years)
	- Expected total number of direct and indirect beneficiaries, disaggregated by gender (reduced vulnerability or increased resilience); - Number of beneficiaries relative to total population, disaggregated by gender (adaptation only)	Total	About 11,300 people (approximately 49% female based on latest available (2013) sex disaggregated data)
		Percentage (%)	100%
Other relevant indicators	<u>Freight costs reduced:</u> The baseline (2016) is \$5,300 per dry goods TEU,¹⁹ and \$7,600 per reefer (refrigerated) container. The target is to reduce these by 15%. <u>Vessel time at port reduced:</u> The baseline (2016) is 14–21 days. The target is 2–3 days.		

Methodology – Mitigation

Without project

Container ships:

- Container ships travel to Nauru.
- After reaching Nauru waters, container ships moor or drift at sea for an average of 20.71²⁰ days while unloading/loading takes place. Container ships use, on average, 20 tons of fuel per day²¹. Hence, each container ship uses 414.2 tons of fuel during unloading/loading.
- Annually, on average, 10 container ships visit Nauru. Hence a total of 4,142 tons of fuel are used by container ships during unloading/loading.
- The CO₂ emission factor (ton of CO₂ per ton of fuel) is 3.114²². Therefore, annual emissions = 12,898 tons CO₂ (4,142 t x 3.114).

Pusher barges²³:

- Pusher barges travel out to the container ships, load/unload, and travel back to port.

¹⁹ Twenty- foot equivalent unit.

²⁰ *Feasibility Study, ADB TA-9009 NAU: Port Development Project, Version 7*, Cardno, 2017

²¹ Data provided by Neptune Shipping company

²² CO₂ emission factor for heavy fuel oil. Source: "Third IMO Greenhouse Gas Study 2014", IMO, 2014.

²³ Data (for all data in this section) provided by Nauru Port Authority (NPA)

- For each container ship, on average, 2 pusher barges work 12/hours per day continuously (unless extreme weather conditions).
- The actual operating time - i.e. time with engine running - is 60% of the time. Hence, for each container ship, during the 20.71 days of unloading/loading, each pusher barge is operating for 149.112 hours on average.
- A pusher barge uses 20 liters of fuel per operating hour.
- Combined, for each container ship, the two pusher barges use 5964.48 liters of fuel during unloading/loading.
- Annually, there are on average 10 container ships, hence pusher barges use 59,644.8 liters of fuel, or 57.199 tons²⁴.
- The CO₂ emission factor is 3.114²⁵. Therefore, annual emissions = 178 tons CO₂ (57.199 t x 3.114).

Fuel ships:

- Fuel ships travel to Nauru.
- After reaching Nauru waters, fuel ships moor or drift at sea for an average of 4.31 days²⁶ while unloading take place. Fuel ships use, on average, 20 tons of fuel per day²⁷. Hence, each fuel ship uses on average 86.2 tons of fuel during unloading.
- Annually, there are on average 5 fuel ships, hence during unloading, 431 tons of fuel are used by fuel ships.
- The CO₂ emission factor is 3.114²⁸. Therefore, annual emissions = 1342 tons CO₂ (431 t x 3.114).

Port facilities:

- The port's current electricity use is 360 kWh/day and 93,600kWh per year (360kwh/day x 5 days per week x 52 weeks per year). All this electricity is diesel generated.
- The diesel fuel consumption for generating electricity is estimated using a factor of 0.29 litres per kWhr.²⁹ Hence, the amount of diesel fuel needed to generate the electricity usage at the port currently is = 93,600 kWhr x 0.29 litres per kWhr = 27,529 litres per year.
- Average CO₂ emission is ~ 2,680 g of CO₂ / litre of diesel burnt.³⁰ Therefore, annual emissions = 2,680 g x 27,529 litres = ~74 tons CO₂.

Annual GHG emissions in the baseline scenario: 12,898 (container ships) + 178 (pusher barges) + 1342 (fuel ships) + 74 (port facilities) = 14,492 tons CO₂.

With Project

Container ships:

- After reaching Nauru waters, container ships will travel directly to port to unload/load.
- Total travel time will be on average 3.75 days³¹.
- Hence, ten ships per year, using 20 tons of fuel per day, will use 750 tons of fuel. This equates to 2336 tons CO₂ per year.

Pusher barges:

²⁴ Assuming the mass density of heavy fuel oil is 0.959 kg/litre.

²⁵ CO₂ emission factor for heavy fuel oil. Source: "Third IMO Greenhouse Gas Study 2014", IMO, 2014.

²⁶ Cardno, 2017

²⁷ Cardno Consulting, personal correspondence confirming fuel ships and container use similar amounts of fuel per day.

²⁸ CO₂ emission factor for heavy fuel oil. Source: "Third IMO Greenhouse Gas Study 2014", IMO, 2014.

²⁹ ADB. 2015. *Proposed Administration of Grant for Nauru Electricity Supply Security and Sustainability Project*. Manila

³⁰ https://people.exeter.ac.uk/TWDavies/energy_conversion/Calculation%20of%20CO2%20emissions%20from%20fuels.htm

³¹ Cardno, 2017

- No pusher barges are to be used to assist loading/unloading.
- One pusher barge will be used to assist the container ships and the fuel ships to dock – assisting a total of 15 ships annually. This will take on average four hours of pusher barge time for each ship (covering both arrival and departure)³².
- A pusher barge uses 20 liters per hour. Hence, annually, for all 15 ships, 1200 liters of fuel be used. This equates to 4 tons CO₂ per year.

Fuel ships:

- After reaching Nauru waters, fuel ships will travel directly to port to unload.
- Total travel time will be on average 1 day³³.
- Five ships per year, using 20 tons of fuel per day, will use 100 tons of fuel. This equates to 311 tons CO₂ per year.

Port facilities:

- The average daily energy usage of the new port when it is eventually at full operational status, is estimated to be 6,777 kWh/day (~ 706 (max demand in kVA) x 0.5 (load factor) x 0.8 (power factor) x 24 hrs/day), and 1,762,020 kWh/year (6,777kWh/day x 5 (days per week) x 52 (weeks per year)).
- The new port will continue to use electricity generated through the diesel generators (though the installation of solar panels on roof space at the new port is being investigated³⁴).
- The diesel fuel consumption for generating electricity is estimated using a factor of 0.24 litres per kWhr.³⁵ The amount of diesel fuel needed to generate the electricity needed at the new port is = 1,762,020 kWhr x 0.24 litres per kWhr = 422,885 litres per year.
- Average CO₂ emission is ~ 2,680 g of CO₂ / litre of diesel burnt. Therefore, annual emissions = 2,680g x 422,885 litres = ~ 1133 tons of CO₂ per year.

Annual GHG emissions in the with project case: 2,336 (container ships) + 4 (pusher barges) + 311 (fuel ships) + 1,133 (port facilities) = 3,784 tons CO₂.

Emission reductions:

Based on the above calculations, summarized in Table 8 below, the annual reduction in emissions will be 10,708 tons of CO₂. Over the lifetime of the project, 50 years, the total reduction will be 535,400 tons of CO₂.

³² Data provided by NPA

³³ Cardno, 2017

³⁴ Note: Upon completion of the project, there will be 1,890 sqm of effective roof space available at the port that could accommodate 260kW of 1,050 solar panels. This would incur capital and operational costs of up to \$1 million and could generate up to 18% of the electricity needs of the port facilities. We will continue to investigate this opportunity in a coordinated manner with the government of Nauru, as they are planning to develop their solar energy capacity and any solar initiatives such as this one will need to be coordinated with its overall renewable energy plan.

³⁵ ADB, Government of Australia and European Union are currently co-financing Nauru Electricity Supply Security and Sustainability Project (see footnote 29) for supply, installation and commissioning of 2 medium speed diesel generators for Nauru Utility Corporation and these generators will be operational by end of December 2017. Upon commissioning, the electricity generation efficiency in Nauru is expected to increase from current 3.4 kWh per liter of diesel consumed to 4.1 kWh per liter.

Table 8: Summary of annual CO₂ emissions under baseline scenario and with project

Sources of CO ₂ emissions	Annual tons CO ₂ emissions – baseline	Annual tons CO ₂ emissions - with project	Difference in annual tons CO ₂ with project
Container ships	12,898	2,336	-10,562
Pusher barges	178	4	-174
Fuel ships	1,342	311	-1,031
Port facilities	74	1,133	+1,059
Total	14,492	3,784	-10,708

Key notes and assumptions:

1. In the with and without project scenarios, travel time to/from Nauru is same for all ships, as is fuel use.
2. It is anticipated that the project outputs will generate additional shipping traffic to Nauru. The forecasts are that there will be, on average, 2 additional container ships and 1 additional fuel ship annually, traveling to Nauru and contributing to GHG emissions. However, there is no baseline scenario counterpart to this increased traffic scenario, and hence so these emissions are not considered in the above calculations.
3. Displacement. There is a risk that the shipping companies will use the ships to travel elsewhere with the time saved, and this may use an equivalent quantity of fuel. Little is known about this possibility, or as to where the resulting emissions be. It is also possible that, globally, shipping companies will simply reduce the size of their fleet given this reduced requirement for shipping.
4. The amount of cargo handling will be the same before and after the project. As such, diesel consumption of loading equipment was not included in the calculation, as it will be cancelled out when comparing the with and without project scenarios.

Methodology - Adaptation

The port is an essential national asset and contributes substantially to all socioeconomic development on Nauru. (i) All the island people are therefore considered beneficiaries, and (ii) the project makes a meaningful contribution to all-round adaptive capacity.

The GCF investment of \$26.91 million will benefit the entire population of about 11,300 people, which amounts to about \$2,381 per beneficiary.

E.2. Paradigm Shift Potential

Degree to which the proposed activity can catalyse impact beyond a one-off project/programme investment

E.2.1. Potential for scaling up and replication (Provide a numerical multiple and supporting rationale)

Potential Pacific Region Paradigm Shift

All 14 Pacific developing member countries and hundreds of small islands in those countries face situations like that of Nauru's. They are dependent on a single port that is small and poorly protected or not protected at all. Unloading and loading operations often take place at sea and are very dangerous. Delays are common. These Pacific Ocean ports also face a similar set of climate and climate change threats:

Winds. Trade winds leading to swells affect port operations, and strong monsoon winds cause delays and damage.

Storm winds and storm sea surges. Storm winds and associated waves cause damage to such onshore and offshore infrastructure as quays, wharfs, breakwaters, cargo sheds, passenger sheds, cranes, mooring sites, and access roads.

Sea level rise. Rising sea levels exacerbate these other effects. Sea level rise may undermine the performance of assets and cause damage on its own.

Extreme precipitation. Extreme rain may contribute to flooding and cause damage.

The Pacific islands and their peoples have much in common with Nauru and its people. They are vulnerable to their climate and to climate change. They are small in area, population, and the size of their economies. They have low trade

volumes, face high construction costs, endure remoteness from supply chains and markets, have only small human resource pools, and face a challenging physical environment.

Climate-resilient ports would greatly contribute to resilience across the region. Until now, however, very little climate change science has been applied for the development of maritime transport, especially in the Pacific. This may be one of the first major port's in the Pacific to consider climate change, and that makes it highly innovative for the Pacific. The approach of collecting and applying climate science to this project can be replicated throughout the Pacific by ADB and other accredited entities and government partners to create a paradigm shift towards climate resilient ports.

Paradigm Shift for Nauru

The current connectivity, through the port, is limited. The current shipping arrangements are ill-suited for economic, social, cultural and political development. All international trade and transport is undermined by the port. In many ways, the port is a bottleneck to connectivity, rather than a facilitator. Climate change will exacerbate this situation, further reducing connectivity, and further constraining sustainable economic, social, cultural and political development.

The project will construct a year-round port with a protected harbour and shore-backed berth for oceangoing vessels. This will enable Nauru and its people to consistently, reliably and affordably connect with the regional and global society and economy. This will allow a much freer two-way flow goods and services, information and ideas, proving a booster to economic, social, cultural and political development. It is anticipated this will be a transformative asset for Nauru, and that it will help to ensure the sustainability of Nauru as an island nation into the future.

E.2.2. Potential for knowledge and learning

Building climate-resilient port infrastructure is a new field in many parts of the Pacific. In the past, ports have often been built on a best-guess basis, with very little understanding of wave climates and ocean dynamics. This project has taken a rigorous, science-based approach to designing the resilient port and to selecting the most appropriate option (Annexes 2B and 2N). The design is compatible with robust, yet simple construction techniques that require low levels of maintenance. This can be a model for all other islands in the Pacific that have small or limited port capacity.

Preparing this project has been a learning process, and the lessons are applicable to many ports on many islands across the Pacific. Through monitoring, awareness raising, exchange visits, and participation in regional knowledge sharing, the project will continue to contribute to regional knowledge development. In the final year of the project, a technical assessment will be carried out by an expert team to review the effectiveness of the project and to capture all lessons.

E.2.3. Contribution to the creation of an enabling environment

Enabling growth through a modern, resilient, and reliable port. Rather than hobbling the economy, which the climate-exposed, costly, and erratic port operations currently do, the modern, resilient, and reliable port will help spur Nauru's economic growth and regional integration. The long-term assurance of a proper, dependable flow at more reasonable costs of imports and exports will provide new lifeblood for socioeconomic development and for an enabling environment that will make new economic activities feasible.

Value addition and increased participation in the fishing industry. More directly, the new port will facilitate the expansion and value addition of the fishing industry. This includes the provision of cold storage, which will provide an incentive for international fishing fleets to use the port.

E.2.4. Contribution to regulatory framework and policies

The project includes a reform component that will align governance of the port authority to ensure long term sustainability of the new infrastructure and the revised services offered by the port, and are in line with international best practices. This will include legislative and tariff reform. See component 2 and Annex 2 M for more details. GCF is not contributing to this component so it is not discussed further here.

E.3. Sustainable Development Potential

Wider benefits and priorities

E.3.1. Environmental, social and economic co-benefits, including gender-sensitive development impact

The project will contribute to the sustainable connectivity of the entire population and economy of Nauru and provide a very broad range of important socioeconomic benefits. By giving Nauru a reliable, efficient, affordable, year-round, climate-sturdy maritime transport facility for the first time in its history, the project will have an impact on virtually all the social and economic aspects of life on the island. The project design will include activities aimed specially at supporting the development of poor and marginalized sections of community, notably women.

Economic Co-Benefits and New Economic Opportunities:

Additional investment and trade diversification. The improved efficiency of the port will enable economic diversification and make all economic investments on Nauru more attractive. A port that is not plagued by constant delays, regular shutdowns, slow and costly at-sea transshipment processes, and high rates of cargo damage will make the supply of food, health, fuel, and potable water more reliable and in some cases, less expensive. This will in turn make Nauru more conducive to economic activity and more attractive to potential economic investment than it is now. Areas where the improved competitiveness that a truly functional port will provide may draw interest from both local and regional investors include tourism, finance, food production, and food processing.

Potential fish industry development. The project will provide facilities that may be used for fish storage and processing. This should attract fishing industry actors and potentially provide local employment as well as government revenue.

Potential to use as a regional port. The improved port facilities at Aiwo will encourage regional neighbouring countries (Federated State of Micronesia, Kiribati and Marshal Islands) to use this port as a transshipment hub for transfer of cargo and containers from oceangoing larger ships. Nauru has a large fuel storage tanks currently outsourced to a private firm. The improved port facilities will encourage this firm to enhance its business and supply fuel to neighbouring counties at a competitive price. These additional activities will enhance port revenue and increase employment opportunities.

Jobs for both men and women. Because more ships and shipping lines will use the proper port that the project will provide, the port will generate more skilled and unskilled jobs for men and women in the shipping industry. The employment generation during project implementation is estimated at (i) 15 internationals and 25 nationals in the construction works and (ii) 12 internationals and 4 nationals in the construction supervision and reforms. Measures will be taken to strive for gender equality in employment at the Port and ensure at least 20% of all new jobs will be for women. NPA is committed to including a large proportion of women (30%) in all training courses. In the medium to long term (up to 2027) there will be no direct incremental employment generation proposed for NPA given the current workforce is in excess of the required upper limit for projected port operations.

Social Co-Benefits:

Better health, educational, and social conditions for the population. By increasing the number of ships reaching Nauru and reducing the time and costs of delivering and unloading cargo, the port will ensure that more essential goods will arrive in the country, including fresher produce and nutritious foods that are often not available. This will

improve the health, educational, and social conditions under which the country's people now live. It will also stimulate local markets for basic imported goods. Since women are generally responsible for selling in local markets, they could generate greater income through this increased activity. The smoother, larger, more reliable flow of goods that the new port and shipping competition will enable is expected to reduce consumer prices, including those for basics. Freight costs, which are passed on to the country's consumers and responsible for the relatively high prices for many goods, are expected to decline on average by 15%. This will benefit the population at large and the poor. Medicines and medical supplies now often delayed or unavailable will be more likely to arrive on time and undamaged. Schools and students will have the educational materials that they sometimes now lack because shipments have been delayed or never arrive.

Reduced risk of injury or death. The offshore unloading operations that the current and baseline scenario port require are often dangerous. They will be eliminated by the new protected port facilities. All loading and unloading will take place using safe, modern procedures directly at the shore-side wharf. The port reform activities under component 2 of the project will improve the port's organization, practices, and worker skills. This will include training and the introduction of procedures to ensure that port operations become safer, efficient and sustainable. The combined impact of these measures will be a significantly decreased risk of injury and death for all workers in the maritime transport sector.

Environmental Co-Benefits:

Hazardous materials removed. The existing land facilities at the port include structures constructed with asbestos materials. These pose a risk to human health and the environment. The project will demolish these structures in a safe process, and the hazardous materials will be removed. Some facilities that have fallen into disrepair and are dangerous will be replaced.

Reduced risk of damage to the environment. The current operational practices cause damage to the ocean floor from groundings, accidents, leaks, and abandoned mooring chains. When visiting ships drift at sea their trailing mooring chains damage the sea floor. The new port will almost eliminate these risks to the maritime environment.

Gender Co-Benefits:

Lower consumer prices and reliable supplies. Women are generally expected to benefit from potential lower consumer price and reliable supplies of essential goods such as medicines and fresh produce due to improved trade from efficient port operations. They would be able to more effectively fulfil their roles in budgeting and providing essential care to the family.

Economic gains. An expected increase in economic activity due to improved trade from more efficient port operations is expected to stimulate employment extending to women. The expected flow of produce to the local market will create more opportunities for them to engage in selling and entrepreneurship. The GAP has outlined specific activities and targets to ensure women's benefits to the project. Key activities to provide economic gains to women include proposed recruitment for skilled and unskilled labour by the contractors, recruitment for PMU staff and recruitment for supervision consultant. The proposed capacity building trainings and preference for women candidates for the PMU are expected to improve women's chances of promotion within NPA and engagement of potential women external candidates. Emphasis on equal pay and equal work opportunities and safety on the job will ensure jobs at the port are decent.

E.4. Needs of the Recipient

Vulnerability and financing needs of the beneficiary country and population

E.4.1. Vulnerability of country and beneficiary groups (Adaptation only)

In the early 1980's Nauru was by some calculations the wealthiest nation on earth. That has changed dramatically. Collapsing global commodity prices have combined with poor development planning to make Nauru one of the world poorest countries in terms of many human development factors, including the following:³⁶

- Life expectancy at birth is estimated to be 60.3 years;
- Infant mortality is estimated to be 33 per 1000 births;
- Youth employment is estimated to be 45%, with overall unemployment at 22.9%.

The latest Household Income and Expenditure Survey in 2013 demonstrated that while the incidence of food poverty was zero, about 24 % of the population and 6.8% of households were living below the basic needs poverty line.³⁷ The survey ranked 7.9% of the country's population and 6.2% of households as "extremely vulnerable." Two particularly vulnerable groups are households headed by women (which experience comparatively high levels of extreme poverty) and widows (60% of whom are classified as poor or vulnerable to varying degrees).

The diet of Nauru's people has deteriorated in recent years, which has led to an epidemic in diet-related disease. An estimated 25% of daily intake consists of rice and raw sugar, and only 30 food items comprise around 85% of daily calories consumed. Farming is practiced only on a very small scale on nutrient-poor land and is energy-dense. Thus, imported processed food that must come in through the country's poorly functioning port has replaced traditional food. This has pushed the national levels of obesity and diabetes (type II) to great heights, and 27% of newborns have a reported low birthweight.³⁸

Climate and Climate Change Vulnerability

A key factor underlying the poverty is the remote and isolated nature of the population and the lack of regional and international. This hinders the arrival of information, goods, services, knowledge and opportunities. Typically, the poor and marginalized members of society are the most affected because they tend to have fewer economic alternatives and lower adaptive capacity.

Currently, climate hazards contribute to port operations being inefficient and even occasionally stopped. Climate change threatens to greatly increase these risks and fully undermine operations of the port. This would greatly increase the remote and isolated nature and lack of connectivity the island and its population, thereby greatly increasing the factors that cause poverty.

Project's Impact on Vulnerability

The project should have a direct impact on climate change vulnerability issues described above, although other measures and reforms may be needed. Specifically, the project should do the following:

- Increase the availability of fresh and nutritious food.
- Increase the availability of fish.
- Lower the costs of all food, health, and education products and services.
- Increase economic opportunities and thereby help people to escape poverty and make healthier living more affordable.

In addition, the project will take proactive measures to reduce social vulnerabilities. It will notably provide training and awareness raising to improve port health and safety conditions.

³⁶ UNDP. About Nauru. <http://www.pacific.undp.org/content/pacific/en/home/countryinfo/nauru.html>

³⁷ Prepared by the Bureau of Statistics, Department of Finance, Nauru.

³⁸ UNICEF. Nauru—Statistics. https://www.unicef.org/infobycountry/nauru_statistics.html

E.4.2. Financial, economic, social and institutional needs

Financial

The project will create a more attractive economic and investment environment. It should indirectly stimulate investment and economic diversification on Nauru, helping to make the economy more robust and sustainable and increasing government revenue.

The project directly addresses port sector governance to put the port management on a firm financial footing and establish a sustainable revenue stream and realistic performance-oriented expenditure plans (see Annex 2F).

Economic and Social

See information provided in sections E.3.1 and E.4.1.

Institutional

Nauru's port sector needs institutional reform (details in Annex 2M). The NPA is currently a hybrid, part state-owned enterprise (SOE) and part government department. The port reform component will support its transition to a full SOE model under government supervision. The project will involve the following actions:

- Prepare and issue regulations to establish new management and financial responsibilities, and establish a reporting structure for NPA.
- Design and oversee internal restructuring of NPA to meet the new regulation.
- Design and support tariff reform to support implementation of the new regulation.
- Develop a new financial management system and asset management system and IT system for NPA.
- Design and undertake a comprehensive training programme for NPA staff to meet new requirements.

E.5. Country Ownership

Beneficiary country (ies) ownership of, and capacity to implement, a funded project or programme

E.5.1. Existence of a national climate strategy and coherence with existing plans and policies, including NAMAs, NAPAs and NAPs

The construction of a climate-resilient port facility has been the centrepiece of the ADB Nauru country program since it was introduced to the program in 2014. The project is fully in line with the following national initiatives.

Nationally Determined Contribution (NDC). Nauru's NDC to the UNFCCC has an adaptation component that prioritises actions that can contribute to development and resilience objectives simultaneously. Priority actions are arranged under the following sectors - water, health, agriculture, energy and food security – and notably, infrastructure and coastal protection are identified as priorities in reducing Nauru's vulnerability to climate change. Further, the resilience of Nauru's food, medicine and energy security situation heavily depends on the stability of shipping services to Nauru. The development of a new port also assists in achieving NDC mitigation goals. As explained in previous sections, ship waiting time, ship traveling time and the use of pusher barges will all be greatly reduced, leading to a significant reduction in the use of fuel, and a significant reduction in the CO₂ emitted. Estimates place these reductions in the range of 588,000 tons over the port's 50-year lifetime. Hence this project aligns well with NDC goals and priorities.

Second National Communication to the United Nations Framework Convention on Climate Change (UNFCCC) 2015. Infrastructure and coastal protection are identified as a key vulnerable sector in the second national communication, which notes that "strategic infrastructure can play an important role in improving economic productivity and/or reducing community vulnerability" and "the sea and air ports are critical assets to ensure supply

of essential goods and services (food, fuel, parts, medical supplies), as well as for export” This project is therefore contributing to the second national communication.

National Sustainable Development Strategy (2005–2025). One of the key strategies of the National Sustainable Development Strategy (NSDS) is “fostering a cultural change towards self-reliance.” The project will contribute to this strategy by creating a port that Nauru can manage on its own and facilitating job creation, inclusive economic growth, and human development across the small country. The port is a basic platform on which development activities can be built. The project directly contributes to the NSDS transport sector goal to “improve transport infrastructure and provide reliable and affordable public transport service” and the associated strategy “to refurbish and develop port infrastructure for vessel handling.”

The Nauru Economic Infrastructure Strategy and Investment Plan. This plan calls for the provision of enhanced infrastructure and utilities services, and the project responds to that priority.

Republic of Nauru Framework for Climate Change Adaptation and Disaster Risk Reduction (RONAdapt).

The project is in line with the framework and contributes especially to one of its priority actions: to reduce coastal risks to key infrastructure and reduce flooding occurrence and intensity.

Further, the Pacific nations have developed an integrated approach to address climate change and disaster risk management. The Framework for Resilient Development in the Pacific (FRDP) has been adopted as an overriding policy and vision document for adaptation in the region. For many nations, including Nauru, the FRDP is at least equally important as the INDC, especially in respect of adaptation. The FRDP emphasizes the importance of transport infrastructure and ensuring that it is climate-resilient. The FRDP also fully integrates disaster management and climate change adaptation. The project has been designed with the FRDP in mind and makes a direct contribution to implementing this framework.

As such, ADB has assisted the development of this project at the request of the Government of Nauru. The NDA has been regularly consulted throughout project development and is supportive of the project and proposal for GCF funding (refer Annex 1A for the no-objective letter).

Other government departments, private companies, land owners and the community have also been consulted throughout preparation of the project (see Annex 2K plus Annex 2I and 2S). Overall there is strong support in Nauru for the project, especially by private, health, education and social sectors of the country. This is reflected in a short video that was prepared on the project (provided in email to GCF).

E.5.2. Capacity of accredited entities and executing entities to deliver

The accredited entity is the ADB. ADB is a multilateral development finance institution that provides loans, grants and technical assistance to its developing member countries. ADB is composed of 67 members, 48 of which are from the Asia and Pacific region. ADB's clients are its member governments, who are also its shareholders. In addition, ADB provides direct assistance to private enterprises of developing member countries through equity investments and loans. In 2016, loan, grant and technical assistance approvals to ADB's developing member countries amounted to \$17.8 billion, and total cofinancing mobilized, with donor support, amounted to \$13.9 billion, bringing total sovereign operations to \$31.7 billion in 2016. Nonsovereign operations for the same year amounted to \$2.5 billion.

ADB and Nauru have been working together since 1991, with approved loans, grants, and technical assistance amounting to over \$25 million. ADB has invested in port projects in several countries across the Pacific, notably the Cook Islands, Papua New Guinea, Samoa, and Tuvalu. Through these investments, the ADB has gained significant experience in the maritime transport sector.

The project sponsor is the project executing entity that is the MOF. See Section C.4 for information on its capacity to execute the project.

E.5.3. Engagement with NDAs, civil society organizations and other relevant stakeholders

Project development

The project has been developed alongside the NPA, while the NDA has been regularly consulted throughout and supports the project (refer Annex 1 for the no-objection letter).

The project development included regular consultations and discussions with a full range of stakeholders. This process included financial and governance related consultations with government agencies such as MOF, MOT, Ministry of RONphos, Department of Lands and Survey, Fisheries and Marine Authority, and Department for Commerce, Industry and Environment; regular ongoing technical discussions with NPA staff; consultation with customary landowners in two official meetings; and consultation with landowners. The latter discussions included consultations on environmental issues. In addition, leaflets and other information material were provided to primary stakeholders, communities, and landowners. Finally, a question-and-answer catalogue was used as part of the social assessment to collect information. Bilateral consultations were also held with the following: Vital Energy Nauru, Nauru Airlines, Ocean to Outback Construction, Canstruct, and Capelle and Partner.

Project Implementation

The NPA will prepare and implement a communication plan for the project and will be overseen by the NPA Board of Directors, aided by Project Steering Committee. Primary and secondary stakeholders will be represented on the Project Steering Committee. NPA will engage a community liaison person during the construction phase.

The communication plan will be based on methods used in similar projects. It will respect local customs and follow local tradition for communication requirements. The first step will be to establish a process for dialogue with various stakeholders. Then, initial consultations will be undertaken to share information about the project, receive information from communities and beneficiaries, and discuss potential impacts and mitigation measures.

Internally, the NPA is committed to ensuring that the regular communication of information about environmental and social issues is delivered to personnel. This is to ensure that they are sufficiently informed and can fulfil their responsibilities in a manner compliant with environmental, health, and safety regulations. All port personnel will be informed of the policies, regulations, procedures, and programs pertinent to their job. Further, under component 2 of the project, information and training will be provided to personnel commensurate with their activities and their responsibilities. Training will be organized and periodic information may be developed to communicate information on progress made by the civil works contractor.

Externally, stakeholders will be engaged to ensure that they are adequately informed about the contractor's progress, particularly about the contractor and the NPA's environmental and social performance. In addition, the NPA will also prepare quarterly and annual environmental and social performance reports, as well as periodic information bulletins to inform stakeholders and the public on environmental and social aspects. The extent of information to be communicated to specific external stakeholder groups (e.g. governmental bodies and monitoring agencies) and the communication methods will be defined by the management depending on the separate requirements of each stakeholder.

The main civil works contractor will prepare a community liaison plan (as part of their bidding documents) that address issues such as noise, dust, employment of unskilled labour, gender inclusiveness, and training of workers..

Grievance Redress Mechanism

The first point of call for all disputes will be the PMU. The PMU will attempt to address all concerns and settle all minor disputes. The NPA/PMU will establish a grievance recourse mechanism to receive and address any concerns related to the project that cannot be resolved or addressed by the PMU. This will be done in coordination with government authorities.

Previous consultations with NPA staff have strongly indicated that the greatest risks to port development are potential disputes over customary landownership and environmental damage. It is therefore expected that the grievance recourse mechanism will have a role in resolving land-related and environmental disputes that arise during project implementation. The PMU will inform all potentially affected persons how to access the grievance recourse mechanism.

NPA will employ a liaison and public relation officer. She or he will be responsible for receiving issues raised by and complaints received from stakeholders and making the initial attempts to resolve them.

E.6. Efficiency and Effectiveness

Economic and, if appropriate, financial soundness of the project/programme

E.6.1. Cost-effectiveness and efficiency

The project will meet Nauru's need for a climate-resilient port in the most efficient way possible. Project identification, design, and selection work has occurred in several stages.

The design of the proposed port draws from the following previous design work and technical studies:

1. Reeves Construction Services and BMT WBM Study (2009)
2. ADB Scoping Study (2009)
3. Nauru Economic Infrastructure Strategy and Investment Plan (2011)
4. World Bank/AusAID Port Infrastructure Priority Needs Assessment (2012)
5. Preparatory Study on the project of Constructing Reef-Edge Quay Wall and Causeway at Aiwo Harbour (2014, JICA)
6. Nauru Port Pre-Feasibility Study, prepared by the Pacific Region Infrastructure Facility (PRIF, 2015)

The project design initially involved a review of the data and findings of all the above and consideration of options presented by the above. Based on the previous work, recent data, and the specific studies supported by this project, six options for port design were established. These six options were considered in detail. Their technical, economic, and climate resilience aspects received particularly close examination. This approach has led to (i) the collection and use of all available data, and (ii) the consideration of all potentially feasible options.

Full details of the options considered and the selection criteria are provided in Sections F1 and F2.

One recommendation emerging from the studies listed above is the need for a reform of port governance and management to address institutional and financial sustainability issues. Hence, an assessment of port reform issues was undertaken (Annex 2M) and used as the basis for identifying a set of port reform interventions.

The total cost of the most favourable options is more than \$65.2 million. Unfortunately, Nauru does not have the resources to finance this port, nor does it have access to commercial lending or other grant financing for such projects. Grant finance from GCF and other partners is essential for this project to proceed.

The entire population of Nauru will benefit from increased resilience because of this project.

Mitigation

The project will reduce CO₂ emissions by 535,400 tons for a GCF investment of \$26.91 million, at a rate of \$50.3 per ton. This compares favourably with other mitigation projects in the region.

The project will reduce CO₂ emissions by 535,400 tons for a total investment of \$65.2 million, at a rate of \$121.8 per ton. This compares favourably with other mitigation projects in the region.

E.6.2. Co-financing, leveraging and mobilized long-term investments (mitigation only)

The financing plan is set out in Table 8.

Table 9: Financing package

Source	Amount (US\$ million)	Percentage
Asian Development Bank	20.70	32%
Government of Australia	13.50	21%
Government of Nauru ³⁹	4.09	6%
Green Climate Fund	26.91	41%
Total Project Cost	65.20⁴⁰	100%

The requested GCF support of \$26.91 million will amount to 41% of the overall \$65.21 million project costs, or 73% of the \$36.80 million of costs eligible for climate change adaptation funding.

E.6.3. Financial viability

Economic Assessment

The baseline scenario has an estimated cost of \$0.18 million and Do-Minimum option \$28.5 million over the project lifetime. The selected option has estimated costs of \$65.2 million over the project lifetime. Table 9 summarizes the projected economic viability of the proposed project, as compared to the baseline.

Table 10: Summary of economic viability

Item	Proposed project
Net present value (US\$)	\$25.31 million
Economic internal rate of return	14.1%
Benefit–cost ratio	1.50

Although the proposed project has far higher initial investment (economic costs), it is favourable to the baseline in terms of net present value (NPV), economic internal rate of return (EIRR), and benefit–cost ratio (BCR). With an EIRR of 14.1% and a cost-benefit ratio (CBR) of 1.50, it is considered a viable investment.

It should be noted that the NPV, EIRR, and BCR were also determined for five other options. Full details are provided in section F1 and in Annex 2D. Section F1 also provides details of the sensitivity analysis.

Financial Assessment

The proposed project has a FIRR of 4.97% and a financial NPV of \$35.8 million (using a financial discount rate equal to the weighted average cost of capital (WACC) of 1.01). Full details of the financial analysis, including sensitivity analysis, are presented in section F1 and in Annex 2F. Even in the worst-case scenario tested under in the sensitivity analysis, the lowered FIRR compares favourably with the estimated WACC of 1.01, thereby substantiating the financial viability of the proposed project.

³⁹ Plus in-kind contribution by Government of Nauru of US\$6.27m.

⁴⁰ This total project cost is exclusive of taxes and duties.

Long-Term Financial Viability

The financial analysis in Annex 2F demonstrates the long-term financial viability of port operations and of the investment in the port. The port reform actions in component 2, notably reforming tariffs and supporting revenue collection, are critical to meeting the long-term financial targets.

E.6.4. Application of best practices

The proposed GCF project incorporates lessons and best practices from several other projects to bring about an impact that is effective, efficient, sustainable, and transformative. These lessons are drawn from ADB's extensive involvement in infrastructure development in the Pacific, including in the maritime transport sector. Recent ADB experience with ports in the region includes operations in the Cook Islands, Papua New Guinea, Samoa and Tuvalu.

Although the best practices adopted in these operations were in some cases established in other parts of the world, they are new to the South Pacific and SIDS. They include (i) the use of locally-applicable wave, climate, and climate change modelling; (ii) the use of physical modelling to determine optimal infrastructure for docking and berthing procedures; and (iii) the establishment of appropriate governance and management structures and procedures.

In the conceptualization and design of this project, an extensive assessment of climate, meteorological, and ocean conditions was undertaken using data from Nauru and similar sites and hindcast methodology (see Annex 2). This contributed to a state-of-the-art understanding of the wave climate in Nauru and of potential changes under climate change. Although not entirely complete due to incomplete data, incomplete understanding of climate-meteorology-ocean dynamics, and uncertainties associated with climate change, this has been the most comprehensive application of such techniques and tools in Nauru and possibly for small ports in the South Pacific. The port options assessment and design used the best available information, including on-site geotechnical information and consideration of local resources (RonPhos 200 T excavator) to provide the best possible solution within the financial and logistical constraints of a remote location.

In addition, a Full Mission Bridge Ship Simulator was used to assess the navigation aspects of options under different meteorological conditions. This was used to develop a realistic assessment of the constructability of the options, given the constraints and conditions in Nauru, using experienced maritime and port engineers in the simulations. These simulations helped in highlighting and examining the potential issues with each option. For example, the simulations examined the challenges of entering the berth, docking, and exiting the berth for every option. The simulations then allowed the development of procedures to deal with the issues. The simulation methodology played a significant role in the option selection process. This is a technological best practice and not often used for small port developments.

Finally, component 2 reform of the NPA is best practice to ensure the institutional and financial sustainability of the project.

E.6.5. Key efficiency and effectiveness indicators

GCF core indicators	Estimated cost per t CO ₂ eq, defined as total investment cost / expected lifetime emission reductions (mitigation only)
	\$45.7 of GCF funds invested per ton of emissions reduced. \$110.8 of all funds invested per ton of emissions reduced.
	Expected volume of finance to be leveraged by the proposed project/programme and as a result of the Fund's financing, disaggregated by public and private sources (mitigation only)
	\$38.29 million from public sources.

Other relevant indicators (e.g. estimated cost per co-benefit generated as a result of the project/programme)	
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F.1. Economic and Financial Analysis

Six alternative approaches to the port were fully assessed. These are:

- **Baseline scenario.** No-project; Zero capital cost (Zero CAPEX)
- **Do-Minimum Option.** Reinstall mooring system and upgrade the existing infrastructure.
- **Option 1.** Build a 120 m long wharf along the edge of the reef connected via a rock-filled causeway to an adjacent harbour.
- **Option 2A.** Build a 30 m wide wharf adjacent to a 50 m wide berth pocket dredged into the reef with a breakwater about 170 m long on the seaward side.
- **Option 2B.** Build a 30 m wide wharf adjacent to a 50 m wide berth pocket dredged into the reef (i.e., option 2A without the breakwater).
- **Option 3A.** Build two berths, one on each side of a 30 m wide wharf. Each berth pocket to be dredged into the reef. The landward side berth would be 30 m wide, and the seaward side berth 50 m wide. Build a breakwater about 209 m long on the seaward side.
- **Option 3B.** Build two berths, one on each side of a 30 m wide wharf. Each berth pocket to be dredged into the reef. The landward side berth would be 30m wide, and the seaward side berth 50m (i.e. option 3A without the breakwater).

The details of the options are described in the feasibility study (Annex 2). All six options were subjected to full economic assessment. The findings are presented in Table 10, and full details are presented in Annex 2D, based on cost estimates provided in Annex 2C.

Table 11: Economic assessment of options

	Baseline (Zero CAPEX)	Do- Minimum	Option 1	Option 2A	Option 2B	Option 3A	Option 3B
Economic costs (US\$ million)	0.18	28.4	71.4	62.5	60.6	78.7	73.0
NPV (US\$ million)	n/a	8.54	4.22	25.31	23.69	14.67	13.85
EIRR	n/a	13.0%	9.8%	14.1%	14.2	11.5%	11.6%
BCR	n/a	1.39	1.07	1.50	1.51	1.24	1.24

The economic analysis was carried out in accordance with the revised ADB's Guidelines for the Economic Analysis of Projects (March 2017). The feasibility of the investments was determined by computing the EIRR of the proposed project using a 9% discount rate complying revised ADB guidelines 2017.

The period of analysis covered 30 years despite the design life of the assets will be 50 years. ADB guidelines on economic analysis of projects recommend 20 - 25 years of project period for the economic assessment. Given the longevity of the assets here, a 30-year project period has been adopted. The assessment periods longer than this is not recommended as (i) forecasting costs and benefits into the future potentially increases the unreliability of the assumptions used and (ii) the contribution to the overall results would typically minor after 30 years at the discounting factors adopted (i.e., 9%). As recommended in the ADB guidelines, an estimate of the residual value of the infrastructure (e.g. the remaining 20 years) has been included in the assessment to capture the difference between the design life and the economic assessment period in the final year of the assessment (i.e. year 30).

The analysis considered the following costs and benefits:

Costs

- Capital costs during construction.

- Annual operation and maintenance (O&M) costs during each year of the project's economic life.
- Periodic maintenance costs to be incurrent at specific points in the economic life.
- Residual value of the project infrastructure remaining at the end of the assumed 30-years economic life.
- Environmental and social costs, such as those for land use compensation.

Benefits

- Cargo and fuel vessel time and cost savings.
- Container rate savings.
- Pusher barge operation costs savings.
- Charter vessel costs savings.
- Port operation savings.
- Climate change adaptation savings

A series of nonquantifiable benefits that were not considered would lead to a stronger justification of the project if included in the analysis results. These include continuity of service, savings on container rates, an emergency response facility, improved access to medical supplies. Details of these and other non-quantifiable benefits are provided in Annex 2D, Section 6.

Options 2A and 2B were the most economically attractive. However, in terms of achieving climate resilience 2A is far superior to 2B. Option 2B doesn't have a breakwater and therefore would endure damage from storms and wave action which is difficult to fully quantify but considered significant enough to make the option unviable.

The robustness of the economic performance of option 2A was assessed through a sensitivity analysis. The analysis looked at sensitivity to adverse changes in estimated costs and benefits and changes in the discount rate. The project remained feasible through all likely scenarios (see Annex 2D).

A financial assessment of this option 2A was then undertaken. In summary, the financial assessment determined that option 2A, the selected option, has a financial internal rate of return (FIRR) of 4.97% and a financial NPV of \$35.8 million. These figures are determined using a financial discount rate equal to the WACC of 1.01. Annex 2F shows full details of the data and assumptions used to project financial costs and benefits. It also provides sensitivity analysis. Annex 2F provides information on the previous financial performance of MOF and the NPA, an assessment of their financial management capacity, an analysis of financial sustainability, and financial indicators.

F.2. Technical Evaluation

All six options were subjected to a detailed technical evaluation. The findings are summarized in Table 11, with details provided in Annexes 2 and 2B.

Table 12: Summary of technical evaluation

Option	General design features	Main advantages	Main disadvantages
Do minimum (baseline scenario)	Mooring system comprising deep sea anchors, chains, connecting plates, shackles and mooring buoys.	• No dredging required. • Low up-front costs. • Limited unexploded ordinance risk.	• Still results in delayed offloading times. • Does not address climate change and will involve increasing periods of inactivity due to high wave activity. • Requires continued maintenance of barges to unload container ships. Per the Government, the maintenance cost for the mooring system is US\$6—7m every 5—6 years.
Reef edge (option 1)	120m long and 30m wide wharf with 2 mooring dolphins.	• Limited dredging required.	• High risk vessel departure manoeuvre with simulation revealing that wind and waves 'push' the vessel onto the reef. A high-powered azimuth tug may be required to assist the ship off the berth

	150m long and 10m wide dual lane causeway. Dredged trim edge of reef to -10m LAT.	- Vessels are less exposed to ocean current forces at berth; - Limited unexploded ordinance risk.	and to safely manoeuvre to sea. There is a high upfront cost for this tug as well as an operational and maintenance cost to consider. - Dredging from a spudded barge (costly and timely) is likely required to excavate rock with Standard Penetration Test (SPT) 'N' values of 50+. - Construction in exposed environment will increase construction cost to account for weather risks. - Vessels are likely to 'beam on' to waves likely resulting in lower berth availability. - Fenders on wharf are unlikely to last more than 5 years and will need replacing. - Questions over the stability and/or interconnectivity of block units. - Questions over the quality of manufacturing on island (QA, materials, handling).
Berth pocket (options 2A, 2B, 3A and 3B)	~160m+ long and 30m wide wharf with adjacent wharf apron / cargo area and 1 mooring dolphin. Breakwater. 25m by 7m ramp to shore. Dredged berth pocket 165m by 50m to -10m LAT (for option 3, second berth pocket is 130m by 30m to -10m LAT.).	- Alignment of berth pocket will result in waves 'head-on' to vessels at berth, minimizing the wave related movements. - Construction can take place 'in the dry' with no or minimal reliance on marine plant. - Most of the reef can be excavated without drill and blast. - Operational advantage in cargo movements.	- Arrival and departure manoeuvres must be undertaken during 'slack tides' due to ocean currents pushing vessels onto the reef and fenders at high speed (will affect berth availability but not significantly). - Current forces up to 1m/s parallel to the reef are expected to cause movement of the future 176 m design vessels. A deep-water dolphin for the fore line will therefore be required in the future.

Based on the findings summarized in Table 11, the baseline scenario and option 1 were not considered technically feasible. The baseline scenario is not at all climate-resilient. Option 1, as demonstrated by the shipping simulations, would lead to delays and danger in ship docking and challenges during offloading. The costs related to this would likely increase with climate change.

Subsequently, option 2A was selected due to its high economic value and climate resilience.

F.3. Environmental, Social Assessment, including Gender Considerations

Environmental Considerations

The project is category B as per ADB's Safeguard Policy Statement (SPS) 2009, and in line with ADB guidelines an initial environmental evaluation (IEE) was undertaken. The IEE concluded that there are no significant identifiable environmental impacts and no reasons to deem the project environmentally sensitive. It further concluded that expected impacts are minor, localized, and acceptable, if the correct mitigation measures are implemented. Next, an environmental management plan (EMP) has been prepared setting out environmental mitigation measures to be incorporated in the design and implementation of the project. The IEE and EMP are in Annex 2G. The IEE notes the following:

- The port is located within the existing land parcel of the port that has already been highly modified and does not support any endemic, endangered, or significant terrestrial biodiversity.

- The port land site does not have any freshwater, forests or agriculture.
- The project includes the coastal foreshore, intertidal reef flat and reef slope. These areas have been highly modified and do not support any shallow water endemic, endangered, or significant marine biodiversity.
- Impacts on the terrestrial and shallow water marine ecosystems and their environments resulting from the pre-construction activities are expected to be very minor.
- The port site does not impact any terrestrial or marine conservation and/or protected area, sites of cultural, customary or heritage significance, nor any national or international endangered or protected species.
- Due diligence and proactive management of all pre-construction activities will ensure limited disturbance to the daily business activities that are to be undertaken within the port and surrounding business and community activities.
- Nauru's laws and regulations associated with labour, employment, and health and safety need to be adopted by the pre-construction contractors and monitored by the NPA.

The EMP in Annex 2G outlines the key measures that will be initiated by the project proponent to manage the project's environmental impacts (see notably Table 3 in Annex 2G).

Social Considerations

The project is category C for involuntary resettlement as per the SPS. There will not be any involuntary resettlement impacts causing physical and economic displacement. Fishing grounds for sale and personal consumption can still be accessed in an alternative fishing ground area north of the wharf.

The project is also category C for Indigenous Peoples as per the SPS. There are no Indigenous People defined as distinct and vulnerable on the island.

A poverty, social and gender assessment was undertaken and a poverty reduction and social strategy prepared (Annex 2I). The main elements of the strategy are as follows:

Land ownership. Before project implementation customary landowners will be consulted and their concerns about environmental issues addressed.

Land transfer. No involuntary resettlement will be necessary for this project. The development of the port will operate in the industrial zone and particularly on parcels of land with existing government lease with customary landowners. Additional land to accommodate the proposed design will be negotiated by the government with the leasing enterprises and its customary owners as it soon to expire. A due diligence report has been prepared to confirm land ownership and ensure proposed arrangements for land lease will be in accordance with pertinent government laws and ADB safeguards principles. Land leases will be prepared and landowners informed before seeking signatories for leases. Adjacent residents and other key stakeholders will be consulted by the implementing agency in the pre-construction phase and during implementation. The contractor will also be required to reconfirm lease agreements and the availability of land or access rights for the project prior to starting construction. Finally, project awareness will be raised using pamphlets and brochures.

Communications. A communication and consultation strategy is being prepared involving primary and secondary stakeholders. The strategy will draw from ADB's Guide to Participation⁴¹ and observe the core principles of effective participation as follows (i) promote accountability and transparency; (ii) allow for participation at all levels particularly the poor and vulnerable; (iii) make participation accessible to all; (iv) value diversity; (v) ensure participation is voluntary; and (vi) encourage stakeholders to create their own ideas and solutions. Information to be reflected in the strategy includes participation objectives; key risks and challenges to achieving the objectives; identifying the main stakeholders with attention given to vulnerable groups such as women and poor; messages for each stakeholder; means of communication which will ensure effective and

⁴¹ Strengthening Participation for Development Results: An Asian Development Bank Guide to Participation: <https://www.adb.org/documents/strengthening-participation-development-results-asian-development-bank-guide-participation>

inclusive participation particularly from the vulnerable; timeline; responsibility and resources required; and a grievance redress mechanism that will be communicated to the communities. Further, the specification of the civil works contract will require the successful contractor to prepare a communications plan and a grievance redress mechanism (in line with the project's stakeholder communication and engagement strategy) that will be communicated to the communities. The communications plan will need to be approved by ADB before the start of any work. In addition, in the bid submission the bidders for the civil works contract will be required to express how they intend to engage the community in the construction works, and this will be assessed as part of the evaluation of their technical proposal.

Civil society organizations (CSOs). CSOs have been involved in consultation processes with residents from the port and Aiwo and with customary landowners from all concerned land lots. CSOs could play a major role in awareness of and training on health issues, and gender and social issues.

Labour market. The project will create short-term jobs for skilled and unskilled labourers during the construction phase (including in simple administrative tasks, heavy lifting work, as cleaners, and for traffic control and hospitality). The employment of women in skilled and unskilled work is proposed. All core international and national labour standards will be complied with. Child labour is not an issue on Nauru.

Gender Considerations

A poverty, social and gender assessment was undertaken and a GAP prepared (Annex 2J). Gender co-benefits are summarised in Section E3 above.

The gender assessment revealed that the key gender issues are in the leadership and decision making areas. The project cannot address these issues specifically but can improve the status of women through some economic gains and special project related activities in consultation participation, employment during implementation, engagement in unskilled labour opportunities, capacity building and training, and awareness building on key social issues. Awareness activities may raise better understanding of different gender needs in transport and the maritime sector, labour laws compliance, equal pay for equal employment and protection of young female workers in the sex industry by awareness programmes. Indirectly the project can encourage the participation of women in discussions of land matters and equal share of land lease payments.

The project will strive to achieve gender equality (i.e. 30% female representation, participation and employment) for activities within its remit, and the GAP includes the following key performance targets and activities:

- Community consultation and participation activities will aim 30% women's involvement.
- At least 30% of female representation in committees for the project if established.
- At least 20% of female recruitment for skilled and unskilled labour by the civil works contractor.
- Organise training for port workers and community members on gender and social issues with 30% of female participants.
- NPA will recruit (additional to the baseline) at least 10% women staff for PMU during construction..

An international social and gender specialist will be recruited, who will assess, consult, train and help manage the implementation of GAP.

F.4. Financial Management and Procurement

Financial management

The approach and risk and mitigation measures related to financial management, including audit requirements, are outlined in section C.4. More information is provided in Annex 2E (financial management assessment).

Procurement

A project procurement risk assessment (PPRA) was carried out in accordance with ADB guidelines. It covered the Department of Finance, MOF, the NPA, and the PMU. The PPRA includes an assessment, a risk mitigation and management plan, and a draft procurement plan. Full details are provided in Annex 2L and some key components outlined below.

Due to a lack of resources, no past involvement in major procurement events and the compulsory outsourcing of procurement (albeit of a minor nature), the procurement risk for the project has been rated as high. As confirmed in the procurement plan, all procurement activities under the project include several hold points that require ADB's prior review and concurrence before proceeding to next step complying to ADB Procurement Guidelines (2015, amended time to time). These hold points include ADB concurrence for (i) bid documents, (ii) any addenda issued during bidding period, (iii) bid evaluation, (iv) any contract negotiations and (v) contract award.

The PPRA includes an initial market survey to identify potential bidders for the civil works. Information to hand suggests the number of international contractors with recent experience in Nauru is limited. However, there may be some strengths and abilities in Nauru that can be built upon and international contractors will be encouraged to recruit local 'contractors' and skilled labor. Contractors will be encouraged to commit to local capacity development.

The project would be implemented under one large contract, covering the excavation of the channel and the ship berth, the concrete wharf, breakwater and the land side buildings, container yard and hard stand areas, mainly because the working area is not large, and it may be difficult to split the works for two different contractors to be working on site at the same time.

The bidding will follow one stage, two envelope bidding procedures following ADB's Procurement Guidelines (2015, as amended from time to time). Draft bid documents for international competitive bidding (ICB) will be prepared based on ADB Standard Bidding Documents (SBD) for Large Works, December 2016.

The consultants will also be recruited and engaged following ADB's Guidelines on the Use of Consultants (2013, as amended from time to time). An international consulting firm will be engaged for construction supervision activities and will be recruited following quality and cost based selection method.

Methods and thresholds. The following methods and process thresholds shall apply to procurement of goods and works, and consulting services:

Table 13: Procurement methods and thresholds

Procurement of goods and works		
Method	Threshold	Comments
International competitive bidding for works	\$100,000 and above	Bidding documents prepared and procurement initiated by PMU. Prior review of the bidding documents by ADB. Bid evaluation and Bid Evaluation Report prepared by PMU. Bid evaluation prior reviewed by ADB before approved by the government tender board.
International competitive bidding for plants and goods	\$100,000 and above	As above.
Shopping for works	Below or equal to \$100,000	As above.
Shopping for goods	Below or equal to \$100,000	As above.
Procurement of consulting services		

	Method	Comments	
	Quality and cost based selection for consulting firm	Consultant selection delegated to ADB using ADB's Consultant Management System (CMS).	

G.1. Risk Assessment Summary

A thorough risk assessment has been completed as part of the feasibility study. The major risks identified for the project relate to technical and operational risks during project construction. They are:

- Difficulty in excavation and dredging the berth pocket.
- Difficulty in driving/boring piles.
- Long supply lines to the site for construction purposes.
- Unexpected rough weather.
- Unexploded ordinances.
- RONPHOS 200T excavator proves unusable.
- Spillage of construction materials.

In addition, the following risks have been identified:

- Complex land tenure procedures cause delays in construction.
- High staff turnover.
- Changes in political leadership lead to delays.
- Spillage of construction materials.

Finally, environmental risks and their management are described in section F.3 above.

G.2. Risk Factors and Mitigation Measures

Selected Risk Factor 1: Difficulty in excavating/dredging the berth pocket

Description	Risk category	Level of impact	Probability of risk occurring
It is proposed to excavate the berth pocket using a land-based excavating plant. Allowance for some drilling and blasting has been included, but if large amounts of hard rock are found, additional drilling and blasting will be required at increased cost and time.	Technical and operational	Medium (5.1-20% of project value)	Low

Mitigation Measure(s)

The geotechnical survey does not indicate large amounts of hard rock. The proposed excavator is considered suitable for completing the required excavation. If drill and blast is required, it is expected to only be for a relatively small quantity of material.

Selected Risk Factor 2: Difficulty in driving/boring piles

Description	Risk category	Level of impact	Probability of risk occurring
The overall program for the marine works package will have a significant impact on cost. The program is dependent on pile driving 'cycle time' assumptions. If equipment breaks down or areas of very hard rock are	Technical and operational	Medium (5.1-20% of project value)	Medium

encountered, this may slow the overall program and result in high labour and equipment hire costs.			
Mitigation Measure(s)			
Part of the risk mitigation will be in tendering the works as lump sum to a main works contractor. This contract seeks to place most of this risk with the Contractor, however, in doing so the Contractors will increase their risk margin and bidding prices. Contractors will be encouraged to have a backup plant and equipment available to minimize very long lead times for equipment otherwise in case of breakdown. Working with contractors familiar with the Pacific and scheduling requirements will also be part of the mitigation.			
Selected Risk Factor 3: Long supply lines to the site for construction purposes			
Description	Risk category	Level of impact	Probability of risk occurring
The remoteness of Nauru means that construction supplies must be ordered and shipped with significant planning. Alternate or additional supplies will have a long lead-time and may delay works incurring costs and time delays.	Technical and operational	Medium (5.1-20% of project value)	Low
Mitigation Measure(s)			
This risk will be largely carried by the contractor. Use of an experienced contractor, preferably with local knowledge and experience, will minimize this risk.			
Selected Risk Factor 4: Unexpected rough weather during construction			
Description	Risk category	Level of impact	Probability of risk occurring
As with any marine construction project, the weather can have a significant adverse impact on progress. The works have been designed to avoid the need for a floating plant which in turn reduces the exposure to weather risks. However, high wave conditions and high sea level may cause problems for construction.	Technical and operational	Low (<5% of project value)	Low
Mitigation Measure(s)			
This risk will be largely carried by the contractor. Use of an experienced contractor, preferably with local knowledge and experience, will minimize this risk. The works should be scheduled and planned to minimize exposure to unexpected or inclement weather conditions; ideally sensitive works would be undertaken during the period from April through to November to avoid the likely periods of highest wave activity on the west coast of Nauru.			
Selected Risk Factor 5: Unexploded ordinance			
Description	Risk category	Level of impact	Probability of risk occurring
A Geophysical Survey for unexploded ordinance has been carried out. A Consolidated Report was prepared and a Risk Assessment (for UXO activities) was conducted based on the results of the Historical Review and the Geophysical Survey. It was determined that there is a high probability that hazardous explosive remnants of war (ERW) will be encountered during the progress of the proposed Nauru Port Development Project.	Technical and operational	Low (<5% of project value)	Medium

Mitigation Measure(s)			
Using the UXO Risk Assessment, a Risk Matrix was prepared to consider the influence of individual project activities on the possible presence of the known munition types. The Risk Matrix then provided the mitigating action/s required in order to conduct the activity safely. Based on the initial finding of the UXO report it was recommended that a qualified Explosive Ordnance Disposal (EOD) Technician should be engaged for intrusive geotechnical activities. With proper procedures and implementation, it is considered that the risk from UXO can be mitigated so that the residual risk is low.			
Selected Risk Factor 6: RONPHOS' 200T Excavator proves unusable			
Description	Risk category	Level of impact	Probability of risk occurring
The project costings assume that a 200T excavator owned by Ronphos, and which is available for the project, is in working order, or can be readily repaired. The machine is currently not in service, but is understood to require relatively minor repairs. If the machine is not serviceable, alternative means of excavating the berth pocket will be required. This may require mobilization of a large machine from contractor's country or within region.	Technical and operational	Medium (5.1-20% of project value)	Low
Mitigation Measure(s)			
This risk will be largely carried by the contractor, which should make its own assessment of the suitability of this plant before pricing the works. Use of an experienced contractor, preferably with local knowledge and experience, will minimize this risk. An alternative suitable machine can be sourced from elsewhere, but presumably at increased cost to the project. Pre-tender investigations by potential contractors should minimize the risk of this factor interrupting the execution of the project.			
Selected Risk Factor 7: Land tenure			
Description	Risk category	Level of impact	Probability of risk occurring
Complex land tenure in project locations can cause delays and limit the successful construction of coastal protection interventions.	Social and environmental	High (>20% of project value)	Medium
Mitigation Measure(s)			
The port operates on several land sections, which are leased by the Government, and adjacent land, needed for the new proposed design, is either customary land without lease or customary land under lease agreement. It is expected that the landowners will allow the construction of the infrastructure. However, given the number of private lands owners involved who all need to endorse the project, this process could create delays in implementation. The project aims to work through the Ronphos to ensure that the communities are part of the decision-making process, thus increasing ownership. The process of obtaining a community endorsement has start during the initial environmental and social impact assessment. Awareness campaigns on coastal protection will improve the understanding of the proposed interventions making sure they are supported and endorsed by the community.			
Selected Risk Factor 8: Staff turnover			
Description	Risk category	Level of impact	Probability of risk occurring
High staff turnover and a limited local human resource base could compromise the project management unit and delay implementation	Technical and operational	Low (<5% of project value)	Medium

Mitigation Measure(s)			
Competitive salaries and incentive packages will be offered. The high priority of the project, and subsequent high profile, should make the positions attractive. Training and other incentives should also help make the positions attractive.			
Selected Risk Factor 9: Political risks			
Description	Risk category	Level of impact	Probability of risk occurring
Political risks: Changing leadership at national and local level result in project delays or a refocus and/or a suspension.	Other	Low (<5% of project value)	Medium
Mitigation Measure(s)			
The probability of a leadership change resulting in a refocus of the project is highly unlikely given that the construction of the port is unequivocally a national priority. However, delays in project decision making is a likely consequence.			
Selected Risk Factor 10: Spillage of construction materials			
Description	Risk category	Level of impact	Probability of risk occurring
Spillage of construction materials. The transport and supply of material, barges, dredges, excavator, truck and any other machinery may have impacts that may arise from accidental spillage of construction materials (e.g. cement); oils and other chemical spills. Further, there may be the release of construction material.	Social and environmental	Low (<5% of project value)	Low
Mitigation Measure(s)			
Through compliance with the environmental and social management plan these risks will be significantly reduced			
Other Potential Risks in the Horizon			
Potential Risks in Price Determination			
The port works proposed will involve one contractor. Although the project budget has been developed with the best available estimates using the industry standards and through consultations with experts who have worked in the region, the overall project budget must be carefully monitored.			
Potential Risks of Community Engagement			
The port works are undoubtedly an urgent priority shared by the Nauru community. This could potentially create impatient demands to see the construction as soon as the project implementation starts. To manage expectations of the community, it will be important to clearly spell out the implementation schedule during the initial consultations, as well as what can be expected and the community engagement requirements.			

H.1. Logic Framework.

H.1.1. Paradigm Shift Objectives and Impacts at the Fund level						
Paradigm shift objectives						
Increased climate-resilient sustainable development	This project delivers essential climate resilience for the people, economy, and society of Nauru by providing climate resilient and efficient port design, construction, and operations.					
Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term (if applicable)	Final	
Fund-level impacts						
A1.0 Increased resilience and enhanced livelihoods of the most vulnerable people, communities and regions.	1.2 Number (percentage) of households, including vulnerable groups such as female-headed/widowed households, adopting a wider variety of livelihood strategies/coping mechanisms. It is proposed that unemployment levels on Nauru be used as a proxy.	National reports	22.9%	To be determined	To be determined	Environmental and social impact assessment is completed without delay; There is a land-use agreement with the landowners.
A3.0 Increased resilience of infrastructure and the built environment to climate change	3.1 Number of new infrastructure projects or physical assets strengthened or constructed to withstand conditions resulting from climate variability and change (e.g. to heat, humidity, wind velocity and floods).	Program reports	Non-resilient port to climate change	0	1. A functioning and climate resilient port	Co-financiers provide indicated support
	3.2 Value of physical assets constructed and/or made more resilient to climate variability and climate change.	Program reports	NA	\$0 (port will still be under construction)	\$62.7 million	Co-financiers provide indicated support
M2.0 Reduced emissions through increased access to low-emission transportation	3.2 Emission levels from vehicles (annual, tons of CO2)	Program reports	14,418	14,418 (no reduction until port is operational)	2,651	No increase in traffic. No displacement of ships to other jurisdictions

H.1.2. Outcomes, Outputs, Activities and Inputs at Project/Programme level						
Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
Project outcome	Outcomes that contribute to Fund-level impacts					
A5.0 Strengthened institutional and regulatory systems for climate-responsive planning and development	5.1 Number and gender sensitivity of policy, laws and sector strategies supported by the Fund	Government gazette notification	0	1 Approved by Parliament by June 2018	1 Approved by Parliament by June 2018	Port reforms supported by Government
A7.0 Strengthened adaptive capacity and reduced exposure to climate risks	Number of days port is closed per year	Annual reports of Nauru Port Authority	2016: 90 days per year	90 (unchanged until port is operating)	Reduced to less than 10 days per year by 2020	Climate change does not lead to unanticipated very extreme events.
M8.0 Increased use of low-carbon transport	Vessel time at port	Annual reports of Nauru Port Authority	2016: 14-21 days	14-21 (unchanged until port is operating)	Reduced to 2-3 days by 2020	Private sector positively responds to development opportunities.
Climate resilience and efficiency of Aiwo port improved	Number of container vessels calls per month	Annual reports of Nauru Port Authority	2016: Less than 1 per month	NA	Increased to more than one call per month by 2020	- Climate change does not lead to unanticipated very extreme events. - Government does not change its strategic priorities contained in the Nauru Port Authority Strategic Plan. - Fuel price increase does not constrain traffic volume. - Private sector positively
	Freight cost	Statistics from shipping operators	2016: USD5300 per dry goods TEU and USD7600 per reefer container	NA	Reduced by 15% by 2020	

						responds to development opportunities. No unexpected factors (e.g. global fuel price rise) affect costs.
Project / programme outputs	Outputs that contribute to outcomes					
Priority port infrastructure constructed or rehabilitated and climate-proofed (Output 1).	Climate-proofed wharf and breakwater constructed	Quarterly and annual PPMS reports of PMU	NA	NA	170m long climate proofed wharf constructed by 2019	- Government provides counterpart contribution on time. - Government budget for operation and maintenance keeps up with needs.
	Climate-proofed access from wharf to shore constructed	Quarterly and annual PPMS reports of PMU	NA	NA	Climate proof access constructed by 2019	
	Port building constructed	Quarterly and annual PPMS reports of PMU	NA	NA	Building constructed by 2019	
	Container storage yard constructed	Quarterly and annual PPMS reports of PMU	NA	NA	Container storage yard constructed by 2019	
	National budget for O&M Plan allocated	Quarterly and annual PPMS report for PMU	2016: Less than A\$100,000	NA	National budget for 2019 allows for full implementation of approved O&M Plan by 2018	
	Public consultations prior to start of civil works completed and participated in by women	Quarterly and annual PPMS reports of PMU	NA	Women's participation rate of 30%	Consultations completed prior to start of civil works, with overall women's participation rate of at least 30%	
Reforms for NPA implemented (Output 2).	Legislation and tariff regulations approved by Parliament	Government gazette notification	NA	NA	Port reform legislation and tariff regulations approved Parliament by December 2018	- Port reforms are supported by the stakeholders - NPA staff are interested in training activities
	Organization structure, business processes, O&M	Quarterly and annual PPMS reports of PMU	NA	NA	100% of approved organization	

	plan and tariff reforms implemented				structure, business processes, O&M Plan and tariff reforms implemented by December 2018	
	Equipment, tools and software procured and new staff recruited	Quarterly and annual PPMS reports of PMU	NA	NA	100% of approved procurement and recruitment completed by December 2018	
	Training plan implemented and participated in by women	Quarterly and annual PPMS reports of PMU	Nil	NA	100% of approved training plan implemented and participated in by at least 10% women	
Efficient project management	Women staff in PMU	Quarterly and annual PPMS reports of PMU	33%	NA	At least 10 % women comprise new PMU staff	
Activities			Description		Inputs	Description
Output 1: Priority port infrastructure constructed or rehabilitated			See Annex 2		See Annex 2	See Annex 2
Output 2.1: Port Reform			See Annex 2		See Annex 2	See Annex 2
The contribution to the ADB Results Framework is provided in Annex 2.						

H.2. Arrangements for Monitoring, Reporting and Evaluation

To be able to measure the mitigative performance (the CO₂ emissions reduced or avoided) of this project, a clear Monitoring and Reporting procedure guiding this project will be agreed between ADB (as the accredited entity) and the Secretariat, prior to the first disbursement. The monitoring and reporting report will be submitted on annually as part of the annual performance report by the ADB to the GCF.

Monitoring, evaluation, and reporting for the project more generally will follow ADB's Evaluation Policy. The PMU will be responsible for project monitoring and reporting. The PMU will prepare biannual reports on progress, on the current level for all indicators, on the implementation challenges and the financial status. The PMU will appoint one staff member as focal point for monitoring and evaluation (M&E).

Overall responsibility for day-to-day project monitoring and implementation rests with the project manager. The project manager will develop quarterly and annual work plans to ensure the efficient implementation of the project. The project manager will inform the implementing agency and the ADB of any delays or difficulties during implementation, including the implementation of the M&E plan, so that the appropriate support and corrective measures can be adopted. The project manager will also ensure that all project staff maintain a high level of transparency, responsibility and accountability in monitoring and reporting project results. The PMU will organize one independent evaluation at the financial close of the program.

Monitoring activities take place as follows:

- A detailed project administration manual will be prepared at the project outset setting out the reporting, monitoring and evaluation activities, responsibilities, and budget.
- ADB will undertake one project review per year to assess progress of project implementation activities, compliance with covenants and project agreements, and to monitor progress in achieving project outputs and agree on any required modifications.
- ADB will undertake a review midway through project implementation or at any time that ADB and the Government consider it necessary. The midterm review mission will (i) review institutional, administrative, organizational, technical, environmental, social, economic, and financial aspects of the project based on the assumptions and risks included in the design and monitoring framework; (ii) review covenants to assess whether they are still relevant or need to be changed, or waived due to changing circumstances; (iii) assess the need to restructure or reformulate the project and the effects of this on the immediate objectives (purpose) and long-term goals of the project; and (iv) update the project's design and monitoring framework if restructuring or reformulation is necessary or its immediate objectives will change.
- Project implementation will be monitored on an ongoing basis by ADB staff in Manila and Sydney.
- Within 6 months of physical completion of the project, the executing entity will submit a project completion report to ADB.

Throughout the life of the project, the MOF, as executing entity, will provide ADB with (i) quarterly progress reports in a format consistent with ADB's project performance reporting system; and (ii) consolidated annual reports including (a) progress achieved by output as measured through the indicator's performance targets, (b) key implementation issues and solutions; (c) updated procurement plan; and (d) updated implementation plan for next 12 months.

I. Supporting Documents for Funding Proposal

- ☒ 1. NDA No-objection Letter
- ☒ 2. Feasibility Study
 - 2A: Concept Drawings
 - 2B: Ship Simulation Report
 - 2C: Cost Estimates
 - 2D: Economic Analysis Report
 - 2E: Financial Management Assessment
 - 2F: Financial Analysis Report
 - 2G: Initial Environmental Examination
 - 2H: Marine Assessment Report
 - 2I: Poverty, Social and Gender Assessment Report
 - 2J: Gender Action Plan
 - 2K: Due Diligence Report - Safeguards
 - 2L: Project Procurement Risk Assessment
 - 2M: Climate Risk and Vulnerability Assessment
 - 2N: MetOcean Criteria Report
 - 2O: Terms of Reference to Advisor for NPA CEO
 - 2P: Port Reform – Organizational Structure Details
 - 2Q: Design & Monitoring Framework (DMF)
 - 2R: Contribution to the ADB Results Framework
 - 2S: Meetings
- ☒ 3. Integrated Financial Model that provides sensitivity analysis of critical elements
- 4. Confirmation letter or letter of commitment for co-financing commitment (Government of Australia, Government of Nauru)
- ☐ Project/Programme Confirmation/Term Sheet (including cost/budget breakdown, disbursement schedule, etc.) – see *the Accreditation Master Agreement, Annex I*
- ☒ 5. Environmental and Social Impact Assessment (ESIA) or Environmental and Social Management Plan
See annexes 2G and 2I.
- ☒ 6. Appraisal Report or Due Diligence Report with recommendations
See annexes 2G and 2I.
- ☐ Evaluation Report of the baseline project (If applicable)
- ☒ 7. Map indicating the location of the project/programme
- ☒ 8. Timetable of project/programme implementation



Republic of Nauru
Department of Foreign Affairs & Trade
Tel: (674) 444 3133 Ext : 267

28 March, 2017

To: The Green Climate Fund ("GCF")

Re: Funding proposal for the GCF by Asian Development Bank regarding Sustainable and Climate Resilient Connectivity for Nauru

Dear Madam, Sir,

We refer to Sustainable and Climate Resilient Connectivity for Nauru (the project) as included in the funding proposal submitted by Asian Development Bank to us on 16 March 2017.

The undersigned is the duly authorized representative of the National Designated Authority of Nauru.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The Government of Nauru has no-objection to the project as included in the funding proposal;
- (b) The project as included in the funding proposal is in conformity with Nauru's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

A handwritten signature in black ink, appearing to read 'Michael Aroi'.

Michael Aroi
Secretary for Foreign Affairs & Trade and ND A



Environmental and social report(s) disclosure

Basic project/programme information	
Project/programme title	Sustainable and Climate Resilient Connectivity for Nauru
Accredited entity	Asian Development Bank
Environmental and social safeguards (ESS) category	Category B

Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	2017-05-09
Language(s) of disclosure	English
Link to disclosure	https://www.adb.org/projects/documents/nau-48480-001-iee https://www.adb.org/projects/documents/nau-48480-001-sprss-0 A simplified impact assessment (ESIA) consistent with the requirements for a category B project is contained in the Initial Environmental Evaluation and the Poverty, Social and Gender Assessment.

Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	2017-05-09
Language(s) of disclosure	English
Link to disclosure	https://www.adb.org/projects/documents/nau-48480-001-iee https://www.adb.org/projects/documents/nau-48480-001-sprss-0 A management plan (ESMP) consistent with the requirements for a category B project is contained in the Initial Environmental Evaluation and the Poverty, Social and Gender Assessment.

Resettlement Action Plan (RAP) (if applicable)	
Date of disclosure on accredited entity's website	Not applicable

Any other relevant ESS reports and/or disclosures (if applicable)	
Description of report/disclosure	Safeguards Due Diligence Report
Date of disclosure on accredited entity's website	2017-05-09
Language(s) of disclosure	English
Link to disclosure	https://www.adb.org/projects/documents/nau-48480-001-sddr

Any other relevant ESS reports and/or disclosures (if applicable)	
Description of report/disclosure	Poverty, Social and Gender Assessment
Date of disclosure on accredited entity's website	2017-05-10
Language(s) of disclosure	English

Link to disclosure	https://www.adb.org/projects/documents/nau-48480-001-sprss-0
Any other relevant ESS reports and/or disclosures (if applicable)	
Description of report/disclosure	Initial Environmental Evaluation
Date of disclosure on accredited entity's website	2017-05-09
Language(s) of disclosure	English
Link to disclosure	https://www.adb.org/projects/documents/nau-48480-001-iee